# 1 Project Charter

## 1.1 Purpose Statement

The University of New South Wales (UNSW) Developmental Solar Farm project aims to provide UNSW with a sustainable and commercial source of renewable energy. The project supports innovation in photovoltaic solar technology, serves as a platform for research and development, and provides UNSW students with opportunities to learn and apply solar technology. The project follows the UNSW 2025 strategy to promote teaching and research in the field of solar energy by increasing the supply of renewable energy.

## 1.2 Requirements

This section lists the key requirements for the project to ensure it meets the University's sustainability goals while supporting teaching and research. In addition, this section highlights the importance of complying with safety regulations, budget and timelines, which together will drive the project's implementation and ensure it is aligned with the UNSW 2025 strategy.
The requirements are as follows:

1. **UNSW 2025 Strategy:** The project will go ahead if it demonstrates alignment with the UNSW 2025 Strategy, Academic Excellence, Innovation and Engagement and Social Impact
2. **Solar Capacity:** The project will initially deliver 20MW of PV cells electrically connected to the grid.
3. **Scalability:** Its electrical reticulation system and electrical switchyard, together with earthworks, will be sized to allow for an additional 10 MW of modules (ie 30 MW total) such that additional developmental PV Modules can be added in the future.
4. **Renewable Energy Supply:**The solar farm will provide a commercial supplement to UNSW's renewable energy supply via the grid.
5. **Education and Research:** The vision of constructing this Developmental Solar Farm is to provide UNSW students the opportunity to study solar energy utilisation and facilitate research and development into advancing PV Solar Energy technology, provides teaching and research opportunities.
6. **Remote Control and Monitoring:** The project provides a control room / monitoring station back on the UNSW campus to allow for remote control and performance monitoring to support learning and research activities.
7. **Documentation:** The project should produce supporting documentation, including R&D documentation, user guides, installation guides, and other relevant materials.
8. **Procurement:** The procurement of all necessary equipment and materials including 20MW of PV Cells and their fixed support frames should be included in the project.
9. **Budget:** The budget of project should be under $25M AUD (fixed).

10. **Timeline:** The project should be completed within 24 months according to the initial plan, however, new project plan calls for shortening project period to 11 months
11. **Safety and Compliance:** Projects should adhere to all safety regulations, environmental compliance requirements and industry standards throughout the development process.
12. **Construction and Commissioning:** The project should complete all construction and commissioning to ensure the solar farm is successfully operated and connected to the grid.

## 1.3 Assumptions

This section describes the key assumptions for project planning and execution.
UNSW has acquired the land for the solar farm project. We assume that the land is suitable for use as a solar farm and no major modifications are required. Therefore, we assume the details listed below.

**Environmental Suitability:**

1. UNSW has acquired the land for the solar farm and the land is suitable for the solar power project without the need for major alterations or adjustments.
2. There is no asbestos or any contaminants on the site.
3. There are no environmental restrictions, such as restrictions on vegetation removal.
4. There is no need to relocate or relocate wildlife.
5. There are no historical artifacts or monuments found on the site.

**Site Conditions:**

1. Existing access to the land is well-established.
2. There are no large bodies of water on the site that require filling or alteration.
3. There are no third-party owned service lines (such as gas pipelines, stormwater pipes, etc.) passing through the site.
4. The site is relatively flat and does not require major land leveling or landfilling.
5. There are no cliffs or major topographic features near the site, and the land continues into a generally flat or hilly terrain.
6. The soil is suitable for cutting and filling, and there are no large areas of rock on or near the surface.
7. No more than 10% of the land is covered with trees.
8. The land is rectangular or trapezoidal in shape, with no complex geometric features.

**Infrastructure Resources:**

1. The site is connected to basic water and electricity services.
2. The site is located near a regional town that can provide the required workforce, living facilities, food and other resources (such as fuel) for the project.

**Monitoring station construction:**

1. The monitoring station on the UNSW campus will require only minimal work, assuming that no major changes to the existing building structure are required, and only existing rooms will need to be modified to accommodate the project needs.

**Funding:**

1. It is assumed that the project will be funded in a timely manner according to the project milestones or phases.
2. The initial budget estimate is that AUD 1 million will be required at the start of the project to carry out the first 3 months of work.

## 1.4 Objectives

The UNSW Development Solar Farm project will deliver a 20MW solar farm and connect it to the grid. The project includes all necessary engineering design, procurement of equipment and materials, civil engineering, and performance testing. The project aims to provide renewable energy supply for UNSW, while supporting future research and development, and reserving the potential for expansion to 30MW.
The key objectives and brief description are as follows:

1. **Alignment with UNSW 2025 Strategy:** The design and implementation of the project is aligned with the UNSW 2025 Strategy, which promotes the university's academic excellence, innovative research and positive impact on society in the field of sustainable energy.
2. **Sustainable Energy Supply:** The establishment of a 20MW photovoltaic solar power generation facility will help UNSW reduce 20% energy costs and 30% carbon emissions, supporting the university's sustainable development goals.
3. **Education Opportunities:** Provide at least 50 students with hands-on experience in studying and understanding solar energy.
4. **Research and Development:** The project will provide infrastructure support for at least 5 related research projects, including cooperation with solar technology optimization, energy storage, smart grid technology and other fields. It is expected to support experimental verification of at least two key research topics within the first year.
5. **Grid Integration:** The power station plans to provide approximately 35,000 MWh of electricity to the grid each year to meet the energy needs of UNSW and surrounding communities.
6. **Scalability and Expansion:** Design the power distribution system and infrastructure to support future expansion to 30MW of power generation capacity, achieving long-term energy supply and project growth potential.
7. **Completion and maintainability:** Ensure that the project is completed as planned within two years, the system maintenance time is controlled to no more than 30 days per year, and the maintenance cost of all equipment is expected to be less than 5% of the annual power generation revenue.
8. **Social Value:** It is planned to create at least 50 direct jobs. In addition, the project will promote community development by prioritizing hiring local labor and partnering with local businesses.
9. **Time and Budget:** Initially, the project should be completed within 2 years, keeping the project cost under AUD 25 million, avoiding budget overruns and schedule delays. However, according to the new requirements of the project, such as adding new equipment and shortening the construction period to 11 months, the new budget will be exceeded, but the optimal budget is always guaranteed.

## 1.5 Definition of Success

The definition of success refers to the criteria for measuring whether a project, program or activity has achieved its intended goals. The definition of success usually includes a series of specific, measurable criteria to determine whether the project has achieved its stated goals as planned, met the expectations of key stakeholders, and was completed within the specified time and budget.
Based on the completion of the requirements in Section 1.2, the weighted analysis is as follows.

| Requirement | Successful Outcome | Importance Weight |
|---|---|---|
| UNSW 2025 Strategy | Alignment with the UNSW 2025 Strategy, Academic Excellence, Innovation and Engagement and Social Impact. | 1 |
| Solar Capacity | The capacity of the solar farm (20 MW of PV cells). | 2 |
| Scalability | Allow for an additional 10 MW of modules (ie 30 MW total) such that additional developmental PV Modules can be added in the future. | 2 |
| Renewable Energy Supply | Provide a commercial supplement to UNSW's renewable energy supply via the grid. | 3 |
| Education and Research | Provide education and research opportunities for at least 50 students or researchers. | 1 |
| Remote Control and Monitoring | Control room / monitoring station allow for remote control and performance monitoring. | 1 |
| Documentation | Supporting documentation, including R&D documentation, user guides, installation guides, and other relevant materials. All required documentation is prepared. | 1 |
| Procurement | Procured materials meet the minimum requirements. | 2 |
| Budget | The initial project budget was A$25m. The latest budget will be exceeded, but it needs to be optimal. | 3 |
| Safety and Compliance | There were no major safety accidents during the implementation process, and all relevant laws and regulations were strictly observed. Various reviews were successfully passed to ensure the safety of staff and environmental friendliness, and a complete emergency response mechanism and regular safety inspections were in place. | 2 |
| Construction and Commissioning | The project should complete all construction and commissioning to ensure the solar farm is successfully operated and connected to the grid. | 1 |

## 1.6 Summary of Stakeholder

| **Stakeholder** | **Benefits** | **Strategy Alignment** |
|---|---|---|

| | | |
|---|---|---|
| UNSW Students/Researchers | It provides research opportunities for solar energy and PV solar energy | Innovation & Engagement; Academic Excellence |
| Grid operator | Provide a more stable and clean energy supply | Social Impact |
| Project investor | The potential for huge commercial returns | commercial potential |
| Local community | Benefit from the clean energy improvements brought about by the project and enhance the environmental quality of the community | Social Impact |
| PV material manufacturer | Economic benefits, while improving their dominant position in the field of photovoltaic | Social Impact / Innovation and Engagement/ commercial potential |
| Construction team | The economic benefits provide jobs while accumulating experience in the PV field | Innovation and Engagement/ commercial potential |
| Project design team | Accumulated valuable engineering experience in the photovoltaic field | Innovation and Engagement/ commercial potential |
| Environmental protection organization | Through the successful implementation of the project, clean energy technology can be promoted to the public and society's attention and support for sustainable energy. | Social Impact / Innovation and Engagement/ |

*Table Stakeholder List*

## 1.7 Summary of Schedule

| **Milestone** | **Milestone Description** | **Target Date** | **Required Preceding Milestone(s)** |
|---|---|---|---|
| 1 | Project Kick-off and Requirements Analysis | Q4 2024 | - |
| 2 | System Design and Preliminary Review | Q4 2024 | 1 |
| 3 | Site Survey and Preparation Begins | Q1 2025 | 2 |
| 4 | Critical Equipment Procurement Completed | Q1 2025 | 2 |

| | | | |
|---|---|---|---|
| 5 | Support Racks and PV Modules Manufacture Begins | Q1 2025 | 2 |
| 6 | Land Clearing and Leveling | Q2 2025 | 3 |
| 7 | Civil Works and Foundation Construction Completed | Q2 2025 | 6 |
| 8 | Support Racks and PV Modules Arrive and Installation Begins | Q3 2025 | 5,7 |
| 9 | Switchyard and Electrical Systems Installation | Q4 2025 | 8 |
| 10 | System Commissioning and Grid Connection Testing | Q1 2026 | 9 |
| 11 | Project Completion and Operational Handover | Q2 2026 | 10 |

*Table Summary schedule List*

## 1.8 Summary of Budget

### 1.8.1 Cost Estimation Strategy

To ensure the accuracy of the project budget, we used the following cost estimation strategies:

**Preliminary Estimation:** First, based on the historical data and experience, we make a preliminary estimate of the major components of the project using a top-down approach to help quickly generate a rough estimate of the overall cost of the project. According to the typical price in Australia is about 1200 AUD/kW × 20,000 kW, the total price is 24 million

**Detailed Estimating:** Based on the preliminary estimate, a Bottom-up approach is used to evaluate specific activities and tasks in the Work Breakdown Structure (WBS) one by one, and to analyze the cost in detail based on the actual work content.

**Non-HR cost:**

**1.** Material and equipment cost estimation: Specific materials and equipment are selected and quantities are determined based on the project design and evaluation. Then analyze and obtain the cost of materials and equipment by referring to the market quotation and supplier information.

**HR cost:**

**2.1** Staff time estimation: Analyze the working time required to complete each task according to the project schedule.

**2.2** Labour Cost Estimation: Estimate the labour cost to complete each task by taking into account the market salary level of different job types.
**Contingency cost:**
**3.** Contingency budget: Based on the risk register, we set aside a portion of the contingency budget to prevent the final budget from exceeding the total budget.

See the detailed report below for the detailed budget calculation

# 1.9 High-Level Risks

## 1.9.1 Risk identification(timing&consequence)

**1. Technical risks**
Equipment failure: Key equipment such as PV modules, inverters, transformers, etc. may fail, resulting in system downtime or interruption of power output.
Technical delays: Deliveries may not be completed on time due to electrical system design or installation problems.
**2. Construction Risks**
Safety accidents: Personnel injuries or equipment accidents may occur at the construction site, which will significantly impact the project schedule and cost.
Construction quality problems: If the construction quality is not up to standard (e.g. unstable bracket installation or irregular cable wiring), it will affect the long-term stability and safety of the system.
**3. Financial Risks**
Cost overruns: e.g. material price increases, construction cost increases or unforeseen additional expenditures that result in total costs exceeding the initial budget.
Financing risk: If there is a delay in financing the project, it will affect the normal progress of the project.
Exchange rate fluctuations: Exchange rate fluctuations will directly affect the purchase cost of imported materials and equipment.
**4. Policy risk**
Policy changes: The government may adjust PV-related subsidies, taxes or regulatory requirements, which will directly affect the project's revenue and compliance.
**5. Climate risk:**
Climatic factors: Extreme weather (e.g., storms, dust storms, etc.) may affect the normal operation of the system and the service life of the equipment.
Construction delays: Bad weather may lead to delays in the material supply chain. And labor shortages may lead to delays in the construction schedule.

**6. Social and community risks**

Community opposition: local residents may oppose the project, especially during the construction period, which may affect the advancement of the project due to issues such as noise, dust and increased road traffic.
Public health and safety issues: the project will need to ensure that it does not impact the quality of life of local residents and that high standards of safety measures are maintained during construction.
**7. Supply chain risks**
Material shortages: During construction, projects may experience a lack of essential materials, particularly if market demand is high.
Transportation risk: Damage and delays may occur during the transportation of equipment, affecting the construction schedule and overall quality of the project.

# 2. Scope Management

## 2.1 Overview of Scope Management (PMBOK Referenced Discussion)

In order to complete this section, we refer to the PMBOK for an explanation of scope management. The PMBOK listed six essential steps to identify and plan the management of scope in Chapter 5:
Plan Scope Management (Chapter 5.1)
Collect Requirements (Chapter 5.2)
Define Scope (Chapter 5.3)
Create WBS (Chapter 5.4)
Validate Scope (Chapter 5.5)
Control Scope (Chapter 5.6)
To plan the scope statement, we used the approach described in Section 5.1.2.3 of the PMBOK, with several online meeting decisions to develop the scope management plan. Through reading the PMBOK Section 5.3.2, we have introduced the project requirements, project deliverables and project details in the article. In order to determine the details of the Scope Statement, we referred to the detailed description of the Project Scope Statement in section 5.3.3.1 of the PMBOK to ensure that the Project Scope Statement contains key information.
In addition, We refer to the PMBOK Section 5.3.3.1 ‘Product Analysis’ for a more detailed breakdown of the project. Then, we used the WBS creation process described in Section 5.4 of the PMBOK to create the WBS. The final WBS was confirmed by voting as described in PMBOK Section 5.2.2.4.

## 2.2 Project Delivery

This project aims to build a small solar power farm covering an area of approximately 35 hectares in New South Wales. This farm is connected to the grid and will be equipped initially with 20 MW of PV cells, possibly expanding to 30 MW in the future. In addition, the

project requires building a control room/monitoring station on the UNSW campus to allow remote control and performance monitoring in support of learning and research activities.

### 2.2.1 Project Design

The project will be designed by civil, mechanical, electrical and energy engineers. The design work includes PV module location design, electrical design, structural and mechanical design, energy transmission design, etc. Design sketches will be submitted to the customer in order for them to have a clear vision of the project process.

### 2.2.2 Project Construction

First of all, the selected land will require the necessary clearing for the construction of the PV farm. The farm will be surrounded by a security fence to ensure the safety of the farm. In addition, several flood emergencies occur in NSW each year. In addition, several flood emergencies occur each year in New South Wales, therefore reasonable drainage design is important to ensure solar farm safety.
Secondly, adequate roads must be constructed within the farm to allow the maintenance and commissioning of solar modules and other facilities. For the future addition of the 10MW solar module area, the road should also be pre-planned and constructed so that it can be put into use quickly in case of future expansion.
Thirdly, the solar modules to be installed and launched in the project need to be installed in accordance with Australian standards to ensure the safety and integrity of the system. At the same time, all solar modules should be ensured to operate at optimal performance and be connected to the grid through specialized inverters to ensure effective energy output and stable power supply.
Finally, during design and construction, it is important that follow the Australian building and electrical standards. This will ensure the reliability and safety of the project and its environmental sustainability.

### 2.2.3 UNSW Monitoring Station

This project will install a monitoring station on the UNSW campus. The monitoring station provides remote control of the solar plant as well as monitoring of the solar farm's performance and data. In order to implement the Renewable Energy Research Funding Scheme, additional instrumentation and control functions need to be installed within the scope of the monitoring room. This creates opportunities for UNSW researchers to work on solar energy.

### 2.2.4 Project Commissioning and Testing

After the project is completed the system is required to be commissioned and tested to confirm that the system is ready for operation, if it is not the technician will be required to commission and re-test the system. The test report will be provided to the customer on delivery of the project to evidence that the solar power farm meets construction standards and is ready for use. Test reports include performance test reports, security inspection reports, operation test reports and stability test reports.

### 2.2.5 Project training documentation

Project documentation includes MonitoringTraining Documents, Equipment Handbooks, Surveillance Training documentation and User Guides. These documents ensure that the operator operates the system correctly and safely.

## 2.3 Constraints

The purpose of constraints is to consider all aspects,such as time,location, finance, and environment, from multiple perspectives. These constraints not only help define the project and limit its scope, but also ensure that the project is completed successfully according to the specified conditions.

The time is to be completed within 11 months, so the project needs to be compressed. The location is on 35 hectares of rural land owned by the University of New South Wales, and this location is fixed and non-negotiable. The budget is the initial budget of 25 million Australian dollars.

## 2.4 Exclusions

The activities and responsibilities that are outside the defined scope of this project are as follows:
1.Further expansion beyond 30MW:Future expansion beyond 30MW is not included in the scope of the project
2.Modifications and designs of grid infrastructure are not considered
3.No mention of operation and maintenance:There is no detailed description of the long-term operation and maintenance activities after the project is completed. Therefore, long-term operation and maintenance are not within the scope of the project definition.
4.Excluding teaching or training activities:While the program aims to support research and development activities, specific research projects or educational courses are not considered.
5.No obligation to dismantle solar farms at the end of their useful lives.
6.This program does not cover changes in energy or environmental regulations that may occur after the project is completed.

## 2.5 Work Breakdown Structure(WBS) - changed

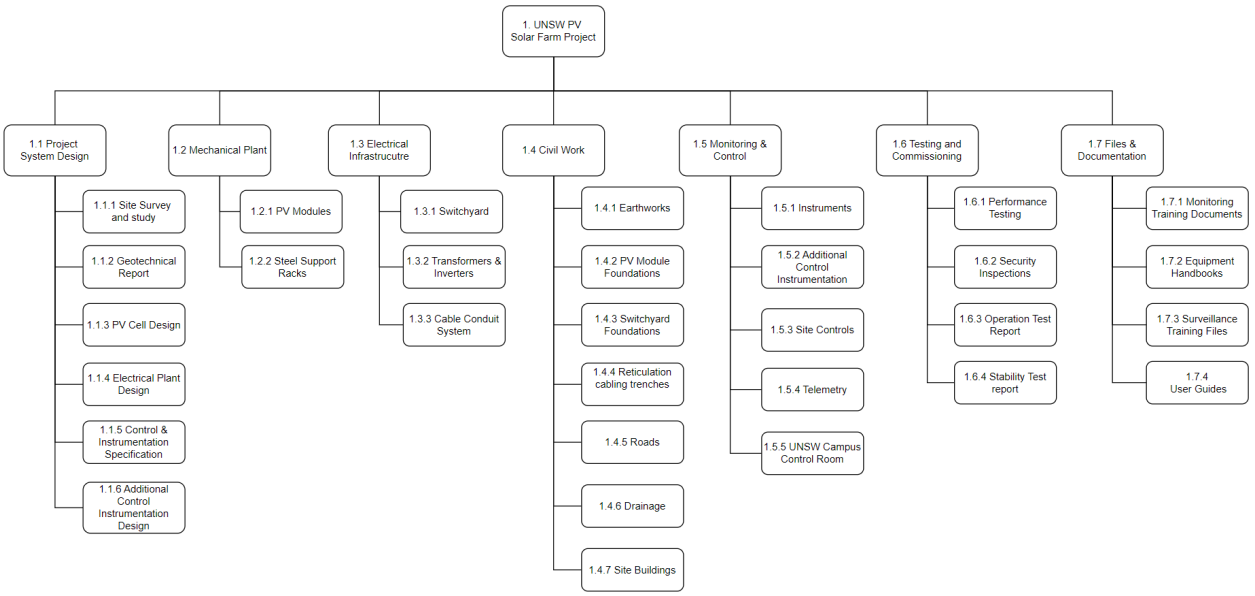



Figure 2.4 Work Breakdown Structure

## 2.6 Scope Statement

The UNSW Solar Farm Development Project includes the design, procurement and construction of a 20 MW photovoltaic solar farm on 35 hectares of land within 11 months, with the possibility of expansion to 30 MW.The project will include the installation of solar panels, support structures, electrical systems, civil works, and a remote monitoring control room on campus. It provides renewable energy for UNSW, supports research and teaching, and tests solar technologies at commercial scale. The scope includes system design, construction, commissioning, training, and project management, aligning with UNSW's 2025 Strategy and utilizing a fixed budget of $25 million.

# 3 Stakeholder Management Plan

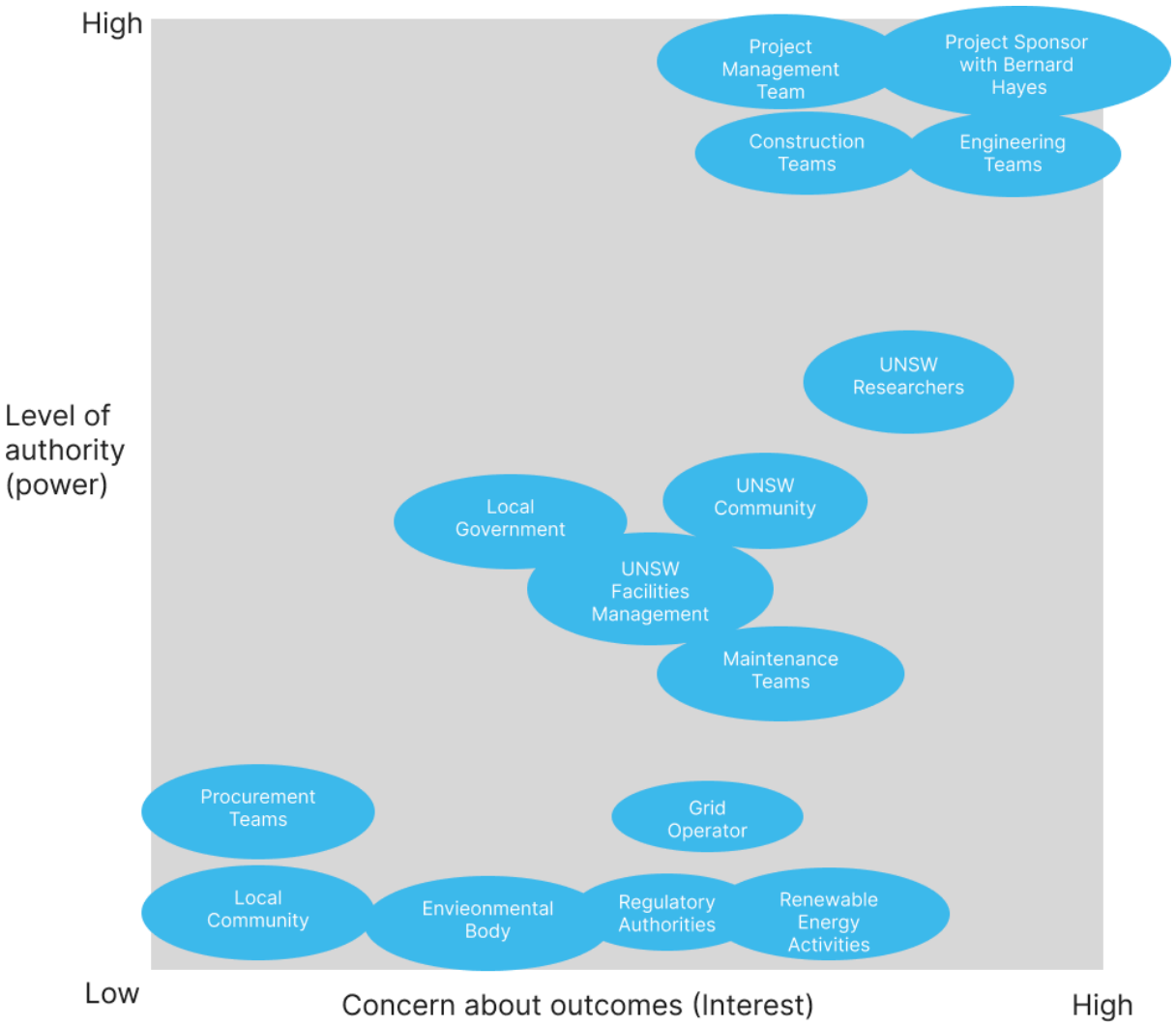
## 3.1 Stakeholder identification - changed

| Stakeholder | Benefits | Strategy Alignment |
|---|---|---|
| UNSW Community | Reduced carbon footprint<br>Self sufficient energy | Social impact<br>Long-term sustainability of compass energy supply |
| UNSW Researchers | Ability to research on solar PV efficiency and technology | Advance knowledge in renewable energy<br>Provides opportunities for research |

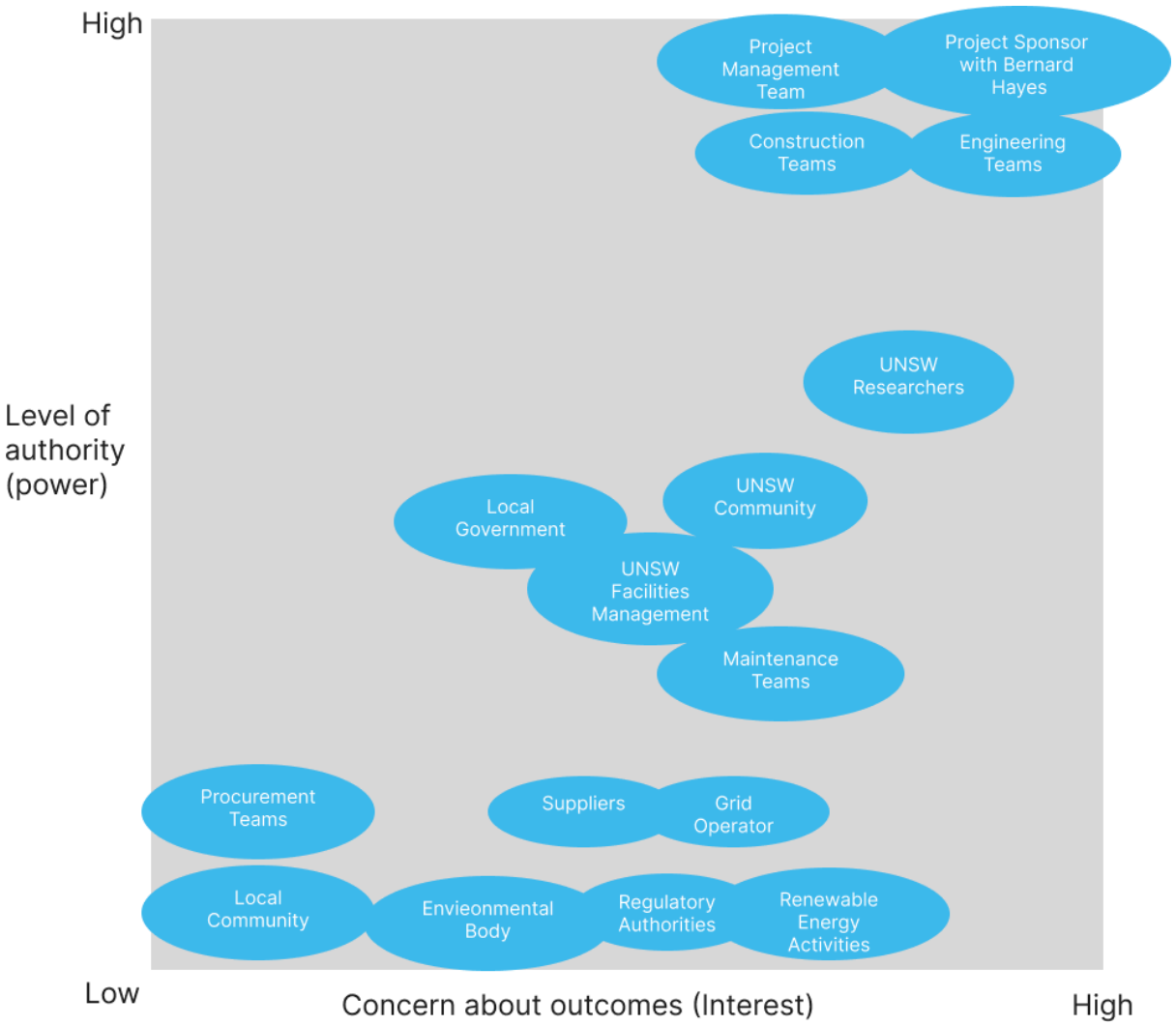
| | | |
|---|---|---|
| UNSW Facilities Management | Efficient energy management system | Campus energy management objectives |
| Project Sponsor with Mr.Bernard Hayes | Enhance credibility by participating in a university project | Promotes renewable technologies |
| Local Government | Economic development through renewable energy projects. | Aligns with local government goals to meet environmental and sustainability mandates |
| Regulatory Authorities | Model for future renewable projects | Aligns with bodies' goals to promote clean energy |
| Grid Operator | Increased renewable energy supply in the grid | Aligns with energy transition goals towards cleaner grid energy |
| Envieonmental Body | Reduced carbon environmental impact | Aligns with goals to reduce fossil fuel reliance |
| Project Management Team | Experience managing a sustainable energy initiative | Project management practices |
| Engineering Teams | Experience in cutting-edge renewable energy technology | Promotes innovation and renewable energy development |
| Procurement Teams | Experience in sustainable supply chain management | Enhance procurement strategies |
| Construction Teams | Experience in renewable energy construction | Supports sustainability related projects |
| Maintenance Teams | Experience in maintaining and operating solar farms | Improve efficiency and reliability in operating large-scale renewable energy systems |
| Local Community | Reduced environmental impact by using clean energy | Aligns with goals of sustainability and environmental protection |
| Renewable Energy Activities | Increased public awareness of the importance of renewable energy | Aligns with the mission of promoting clean energy solutions and reducing the carbon footprint |
| Suppliers | Profit from the project and opportunity to showcase reliability | Aligns with goals to increase reputation |

## 3.2 Stakeholder assessment

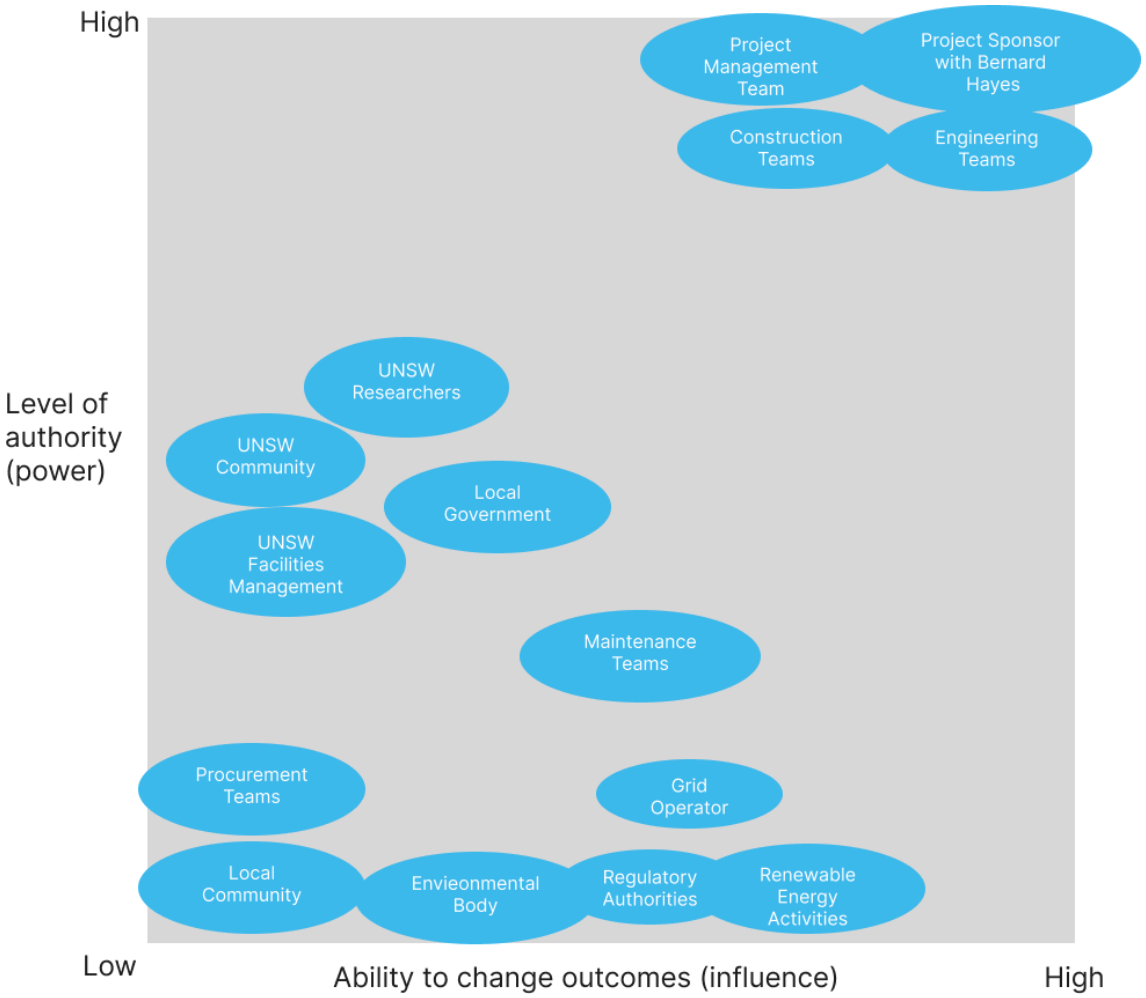
### 3.2.1 Power interest grid - original



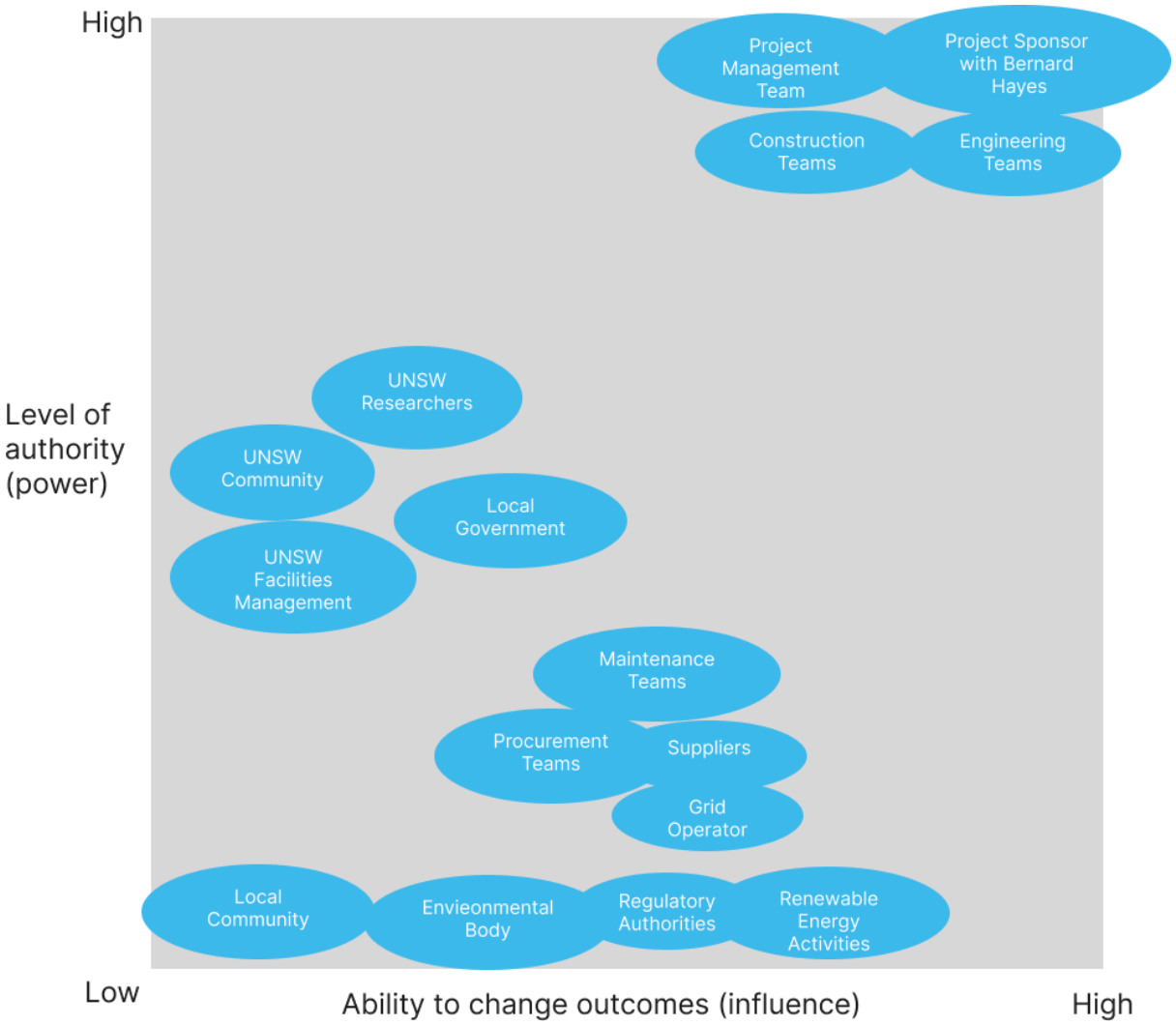

### 3.2.1 Power interest grid - New


High
Level of authority (power)
Low
Concern about outcomes (Interest)
High
Project Management Team
Project Sponsor with Bernard Hayes
Construction Teams
Engineering Teams
UNSW Researchers
Local Government
UNSW Community
UNSW Facilities Management
Maintenance Teams
Procurement Teams
Suppliers
Grid Operator
Local Community
Envieonmental Body
Regulatory Authorities
Renewable Energy Activities

### 3.2.2 Power influence grid - original



### 3.2.2 Power influence grid - New




## 3.3 Stakeholder engagement

| Category | Name | Unaware | Resistant | Neutral | Supportive | Leading |
|---|---|---|---|---|---|---|
| UNSW | UNSW Community | C | | D | | |
| | UNSW Researchers | | | C | D | |
| | UNSW Facilities Management | | | C | | D |
| | Project Sponsor | | | | C | D |

| | | | | | | |
|---|---|---|---|---|---|---|
| | with Mr.Bernard Hayes | | | | | |
| Corporate Bodies and Regulators | Local Govenment | C | | | D | |
| | Regulatory Authorities | C | | | D | |
| | Grid Operator | C | | | D | |
| | Envieonmental Body | | | C | D | |
| General Public | Local Community | C | | D | | |
| | Renewable Energy Activities | | C | | D | |
| Project Teams | Project Management Team | | | C | | D |
| | Engineering Teams | | | | CD | |
| | Procurement Teams | | | | CD | |
| | Construction Teams | | | | CD | |
| | Maintenance Teams | | | | CD | |

C=Current position
D=Desired position

## 3.4 Detailed Stakeholder Management Strategy - changed

### 3.4.1 UNSW

| Identified Stakeholder | Influence | Interest | Management strategy |
|---|---|---|---|
| UNSW Community | Medium Low | Medium High | Inform via general communications and |

| | | | |
|---|---|---|---|
| | | | monitor the general perception of the Project |
| UNSW Researchers | Medium Low | High | Inform via general communications and monitor the general perception of the Project |
| UNSW Facilities Management | Medium Low | Medium High | Inform via general communications and monitor the general perception of the Project |
| Project Sponsor with Mr.Bernard Hayes | High | High | Progress reports and daily project health status |

### 3.4.2 Corporate Bodies and Regulators

| Identified Stakeholder | Influence | Interest | Management strategy |
|---|---|---|---|
| Local Govenment | Medium | Medium | Keep informed and engage on their interest areas |
| Regulatory Authorities | Medium High | Medium High | Keep informed and engage on their interest areas |
| Grid Operator | Medium High | Medium High | Keep informed and engage on their interest areas |
| Envieonmental Body | Medium | Medium | Keep informed and engage on their interest areas |

### 3.4.3 General Public

| Identified Stakeholder | Influence | Interest | Management strategy |
|---|---|---|---|
| Local Community | Low | Medium | Keep informed and engage on their |

| | | | |
|---|---|---|---|
| | | | interest areas |
| Renewable Energy Activities | High | High | Keep informed and engage on their interest areas |

### 3.4.4 Project Construction and Operation Team - changed

| Identified Stakeholder | Influence | Interest | Management strategy |
|---|---|---|---|
| Project Management Team | High | High | Align goals and understanding |
| Engineering Teams | High | High | Regular tracking WP status |
| Procurement Teams | High | High | Collect all the required objects from each department |
| Construction Teams | High | High | Regular construction by all the on-site workers |
| Maintenance Teams | Medium | Medium | Regular inspection of each device |
| Suppliers | Medium | Medium | Close oversight with regular progress tracking |

## 3.5 PMBOK Referenced PM Methods Used in Stakeholder Management – Discussion (Overview of Stakeholder Management)

The PMBOK listed four essential steps to identify and plan the management of stakeholders in section 13:
identifying stakeholders (section 13.1)
plan stakeholder engagement (section 13.2)
manage stakeholder engagement (section 13.3)
monitior stakeholder engament (section 13.4)

**Identifying Stakeholders**
PMBOK provided some useful methods to identify stakeholders, such as expert judgment and data gathering. Our team has identified 15 stakeholders using our expert judgment and

data gathering techniques like brainstorming and surveys and the we made them into a stakeholder list.
PMBOK also mentioned that Stakeholder update is a dynamic process in project management that involves the continuous updating of project management plans and documents. As a project progresses, new stakeholders may be identified or new information may be acquired, resulting in change requests. These changes may affect the management plan in terms of requirements, communication, risk, and stakeholder engagement. At the same time, project documents such as assumption logs, issue logs, and risk registers need to be updated accordingly.

**Plan Stakeholder Engagement**
The focus of the planning stakeholder engagement is to create a comprehensive stakeholder engagement plan that identifies the best way and frequency to communicate with each stakeholder or group.
PMBOK provided some similar tools and techniques to plan stakeholder management. By brainstorming and with the help of the expert judgment, we get a stakeholder engagement matrix from the data of project documents and stakeholder identification list.

**Manage Stakeholder Engagement**
Effective stakeholder engagement management enhances project support, reduces resistance and increases the likelihood of project success. It requires project teams to have good communication skills, be flexible in responding to change, and be able to balance the needs of different stakeholders.
The project team should stay in contact with the stakeholders throughout the life of the project to manage stakeholder engagement. This can be achieved by following the stakeholder engagement plan.

**Monitoring Stakeholder Engagement**
Monitor Stakeholder Engagement is a process that occurs throughout a project with the primary purpose of monitoring stakeholder relationships and adjusting engagement strategies. The core strength of this process is the ability to maintain or improve the efficiency and effectiveness of stakeholder engagement activities as the program evolves and circumstances change. It involves modifying engagement strategies and programs to ensure continued effective stakeholder engagement. The process includes specific inputs, tools and techniques, and outputs.

# 4. Schedule Management

## 4.1 Overview of Schedule Management (PMBOK Referenced Discussion)

In Chapter 6 of the PMBOK guide, titled Project Schedule Management, it outlines the processes necessary to manage the timely completion of a project. The key steps include:

1. Plan Schedule Management: Establishes policies and procedures for planning, developing, and controlling the project schedule.

2. Define Activities: Involves identifying and documenting the specific tasks necessary to deliver the project's objectives.
3. Sequence Activities: This step defines the relationships and logical order of project tasks.
4. Estimate Activity Durations: Involves predicting the time required for each activity based on resource availability.
5. Develop Schedule: Focuses on creating a schedule model that can be used to execute and monitor project progress.
6. Control Schedule: Continuously monitoring the project schedule to manage any necessary changes or updates to maintain timelines.

In order to express schedule management more clearly, we decided to adopt the presentation of Activity List and Network Diagram in Figure 6-2 of the PMBOK, with reference to the examples in 6-3, Figure 6-15, and Figure 6-16, is to better show the overall time Layout.These steps ensure a structured approach to scheduling and controlling project timelines, incorporating methodologies like the Critical Path Method (CPM) and agile scheduling techniques.As for the choice of the project schedule form, we finally voted to confirm Milestone Charts, referring to the example given in 6-21, to complete the form

## 4.2 Activity List

The Activity List is shown in Table 4.2.1 below

| WBS | Working (work package) | | Activity | |
|---|---|---|---|---|
| 1.1 Project System Design | Code | Description | Code | Description |
| | 1.1.1 | Site Survey and Study | 1.1.1.1 | Preliminary site inspection |
| | | | 1.1.1.2 | Investigation and research execution |
| | | | 1.1.1.3 | Data collection and analysis |
| | | | 1.1.1.4 | Stage meeting |
| | 1.1.2 | Geotechnical Report | 1.1.2.1 | Preparation before geological survey |
| | | | 1.1.2.2 | Site drilling and soil sampling |
| | | | 1.1.2.3 | Sample testing and analysis |
| | | | 1.1.2.4 | Data collection and analysis |
| | | | 1.2.4.5 | Summary report of geotechnical engineering |
| | 1.1.3 | PV Cell Design | 1.1.3.1 | Requirements and parameter determination |

| | | | | |
|---|---|---|---|---|
| | | | 1.1.3.2 | Preliminary design and simulation |
| | | | 1.1.3.3 | PV Cell selection and procurement |
| | | | 1.1.3.4 | Steel Support Racks selection and procurement |
| | 1.1.4 | Electrical Plant Design | 1.1.4.1 | Cabling layout and Switchyard design |
| | | | 1.1.4.2 | Inverters arrangement design |
| | | | 1.1.4.3 | Connection to grid design |
| | | | 1.1.4.4 | Final design confirmation |
| | 1.1.5 | Control & Instrumentation Specification | 1.1.5.1 | Requirements analysis |
| | | | 1.1.5.2 | Monitoring Station Design |
| | | | 1.1.5.3 | Control System Design |
| | | | 1.1.5.4 | All Designs Technical review |
| | 1.1.3 | Additional Control Instrumentation Design | 1.1.6.1 | Design Additional Control Instrumentation |
| 1.2 Mechanical Plant | 1.2.1 | PV Modules | 1.2.1.1 | Manufacturing |
| | | | 1.2.1.2 | Transportation |
| | 1.2.2 | Steel Support Racks | 1.2.2.1 | Manufacturing |
| | | | 1.2.2.2 | Transportation |
| 1.3 Electrical Infrastructure | 1.3.1 | Switchyard | 1.3.1.1 | Switchyard Materials Purchasing |
| | | | 1.3.1.2 | Switchyard Materials Transportation |
| | 1.3.2 | Transformers & Inverters | 1.3.2.1 | Transformers & Inverters Purchasing |
| | | | 1.3.2.2 | Transformers & Inverters Transportation |
| | 1.3.3 | Cable Conduit System | 1.3.3.1 | Material |

| | | | | |
|---|---|---|---|---|
| | | | | Purchasing |
| | | | 1.3.3.2 | Material Transportation |
| 1.4 Civil work | 1.4.1 | Earthworks | 1.4.1.1 | Land clearing |
| | | | 1.4.1.2 | Land Leveling |
| | 1.4.2 | PV Module Foundations | 1.4.2.1 | Steel Support Racks Installation |
| | | | 1.4.2.2 | PV Modules Installation |
| | 1.4.3 | Switchyard Foundations | 1.4.3.1 | Switchyard Construction |
| | 1.4.4 | Reticulation Cabling trenches | 1.4.4.1 | Cable Conduit Laying |
| | 1.4.5 | Roads | 1.4.5.1 | Materials Purchasing |
| | | | 1.4.5.2 | Materials Transportation |
| | | | 1.4.5.3 | Road building |
| | 1.4.6 | Drainage | 1.4.6.1 | Materials Purchasing |
| | | | 1.4.6.2 | Materials Transportation |
| | | | 1.4.6.3 | Materials Installation |
| | 1.4.7 | Site Buildings | 1.4.7.1 | Materials Purchasing |
| | | | 1.4.7.2 | Materials Transportation |
| | | | 1.4.7.3 | Construction |
| 1.5 Monitoring & Control | 1.5.1 | Instruments | 1.5.1.1 | Instruments Purchasing |
| | | | 1.5.1.2 | Instruments Transportation |
| | | | 1.5.1.3 | Instruments Installation |
| | 1.5.2 | Additional Control Instrumentation | 1.5.2.1 | Instruments Purchasing |
| | | | 1.5.2.2 | Instruments Transportation |
| | | | 1.5.2.3 | Instruments Installation |

| | | | | |
|---|---|---|---|---|
| | 1.5.3 | Site Controls | 1.5.3.1 | Control System Installation |
| | 1.5.4 | Telemetry | 1.5.4.1 | Software Installation |
| | | | 1.5.4.2 | System Integration |
| | 1.5.5 | UNSW Campus Control Room | 1.5.5.1 | Select a suitable room |
| 1.6 Testing and Commissioning | 1.6.1 | PerformanceTesting | 1.6.1.1 | Performance Testing |
| | | | 1.6.1.2 | Reporting |
| | 1.6.2 | Security Inspections | 1.6.2.1 | Security testing |
| | | | 1.6.2.2 | Reporting |
| | 1.6.3 | Operation Test Report | 1.6.3.1 | Operation Testing |
| | | | 1.6.3.2 | Reporting |
| | 1.6.4 | Stability Test report | 1.6.4.1 | Stability Testing |
| | | | 1.6.4.2 | Reporting |
| 1.7 Files & Documentation | 1.7.1 | Monitoring Training Documents | 1.7.1.1 | Develop documentation |
| | 1.7.2 | Equipment Handbooks | 1.7.2.1 | Develop documentation |
| | 1.7.3 | Surveillance Training Files | 1.7.3.1 | Develop documentation |
| | 1.7.4 | User Guides | 1.7.4.1 | Develop documentation |

Table 4.2.1. Activity List

## 4.3 Milestone List

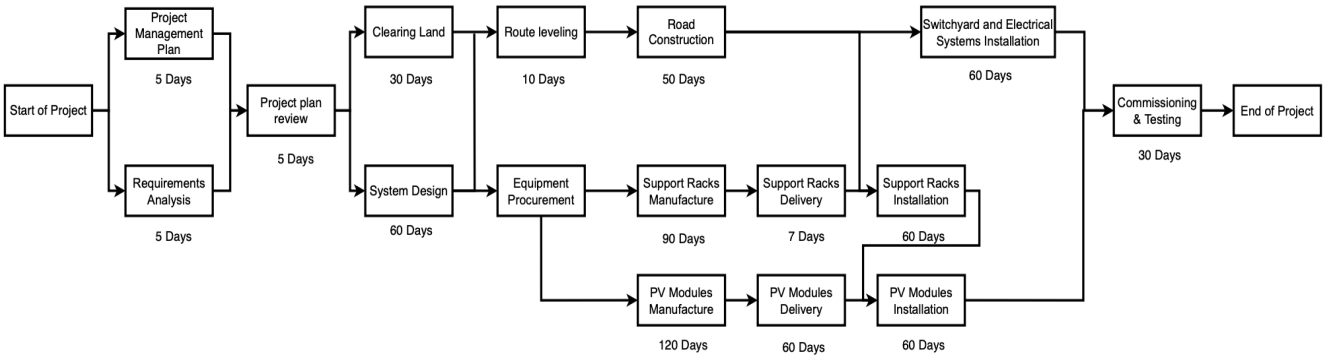
The Milestone List is shown in Table 4.2.2 below

| Milestone | Milestone Description | Target Date | Required Preceding Milestone(s) |
|---|---|---|---|
| 1 | Project Kick-off and Requirements Analysis | Q4 2024 | - |
| 2 | System Design and Preliminary Review | Q4 2024 | 1 |
| 3 | Site Survey and Preparation Begins | Q1 2025 | 2 |
| 4 | Critical Equipment Procurement Completed | Q1 2025 | 2 |
| 5 | Support Racks and PV Modules Manufacturer Begins | Q1 2025 | 2 |

| | | | |
|---|---|---|---|
| 6 | Land Clearing and Leveling | Q2 2025 | 3 |
| 7 | Civil Works and Foundation Construction Completed | Q2 2025 | 6 |
| 8 | Support Racks and PV Modules Arrive and Installation Begins | Q3 2025 | 5,7 |
| 9 | Switchyard and Electrical Systems Installation | Q4 2025 | 8 |
| 10 | System Commissioning and Grid Connection Testing | Q1 2026 | 9 |
| 11 | Project Completion and Operational Handover | Q2 2026 | 10 |

Table 4.2.2. Milestone List

# 4.4 Precedence Diagram




# 4.5 Gantt Chart - changed

*Figure 4.5 Gantt Chart*

## 4.6 Network Diagram - changed

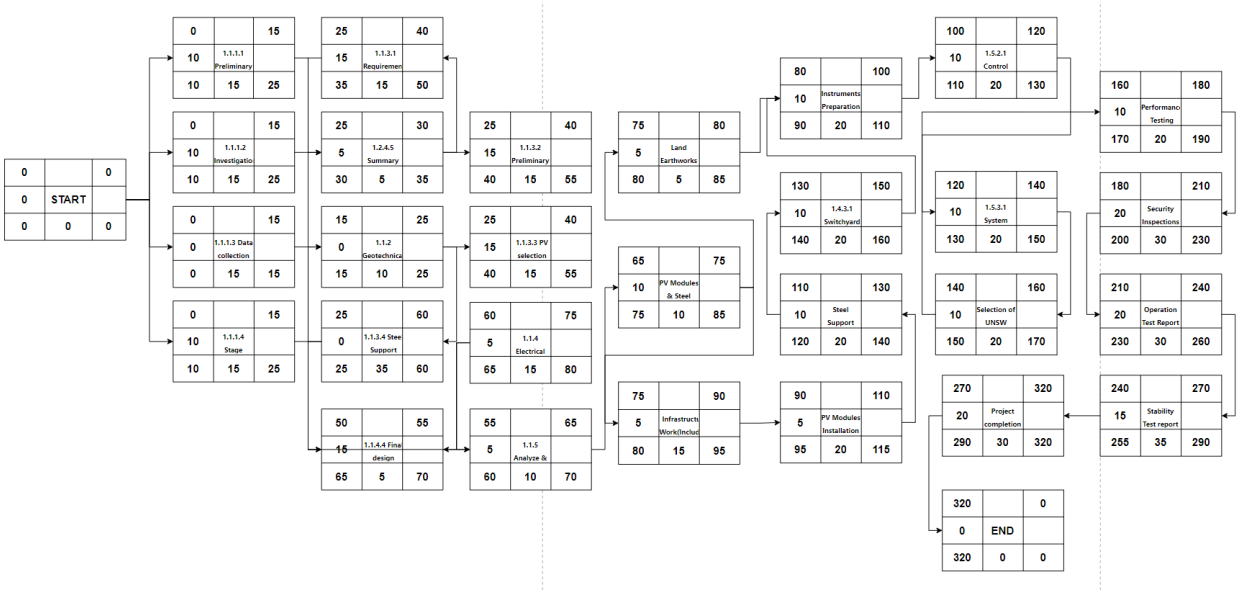



*Figure 4.6 Network Diagram*

# 5. Budget Management

## 5.1 Overview of Budget Management (PMBOK Referenced Discussion)

In Chapter 7 of PMBOK, it is mentioned that project cost management mainly includes planning, estimating, budgeting, financing, funding, managing, and controlling costs.

The core concept of project cost management is to focus on the cost of the resources required to complete project activities and consider the impact of project decisions on the recurring costs of subsequent use, maintenance, and support. For example, reducing the number of design reviews can reduce project costs, but may increase the operating costs of the product. Another key point is to recognise that different stakeholders measure project costs differently at different points in time, such as when a procurement decision is made, when an order is placed, or when the item is delivered. Project cost management also involves predicting and analyzing the financial performance of project products, especially in capital facility projects, which may include techniques such as return on investment, discounted cash flow, and payback period analysis.

The project Budget(Cost) Management processes are:

1. **Plan Cost Management**—The process of defining how the project costs will be estimated, budgeted, managed, monitored, and controlled.
2. **Estimate Costs**—The process of developing an approximation of the monetary resources needed to complete project work.

3. **Determine Budget**—The process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline.
4. **Control Costs**—The process of monitoring the status of the project to update the project costs and manage changes to the cost baseline.

In PMBOK Sections 7.1, 7.2, 7.3 and 7.4, there are details on how to complete each process of Budget(Cost) Management. Section 7.2.2 gives the tools and techniques to help to calculate the budget for each part of the project.

We need to keep track of the timing and cost of the project. Because during the project, we may encounter lots of emergencies, so we need to respond and adjust accordingly. This is related to the changes in the entire process, and we need to keep in close contact with stakeholders during this process to ensure the smoothness and successful project.

For more in-depth modifications, you can refer to PMBOK for further information.

## 5.2 Budget Estimation

### 5.2.1 Cost Estimation Strategy

1. Prepare preliminary estimates for each component in the WBS based on historical information (top-down approach, parametric estimating).
2. Based on the scope of work, use the preliminary estimates as a guide to prepare detailed estimates for each component in the WBS (bottom-up approach)
    a. Estimate the quantity of materials required based on the scope of work
    b. Estimate material costs based on available online information and market quotes
    c. Estimate required employee hours based on the schedule
    d. Estimate employee costs based on the experience level of the required workers and market wage rates

### 5.2.2 Top-Down Approach

Initially, we used a top-down approach to estimate the total cost of the project. We based our estimate on a parameter of AUD 1,200 per kilowatt as mentioned in the project brief.

It is important to note that the rate of AUD 1,200 per kilowatt is volatile and may change due to fluctuations in material futures prices or currency exchange rates. However, with the development of photovoltaic technology, this figure will show a downward trend overall, and this cost may be further reduced in the future.

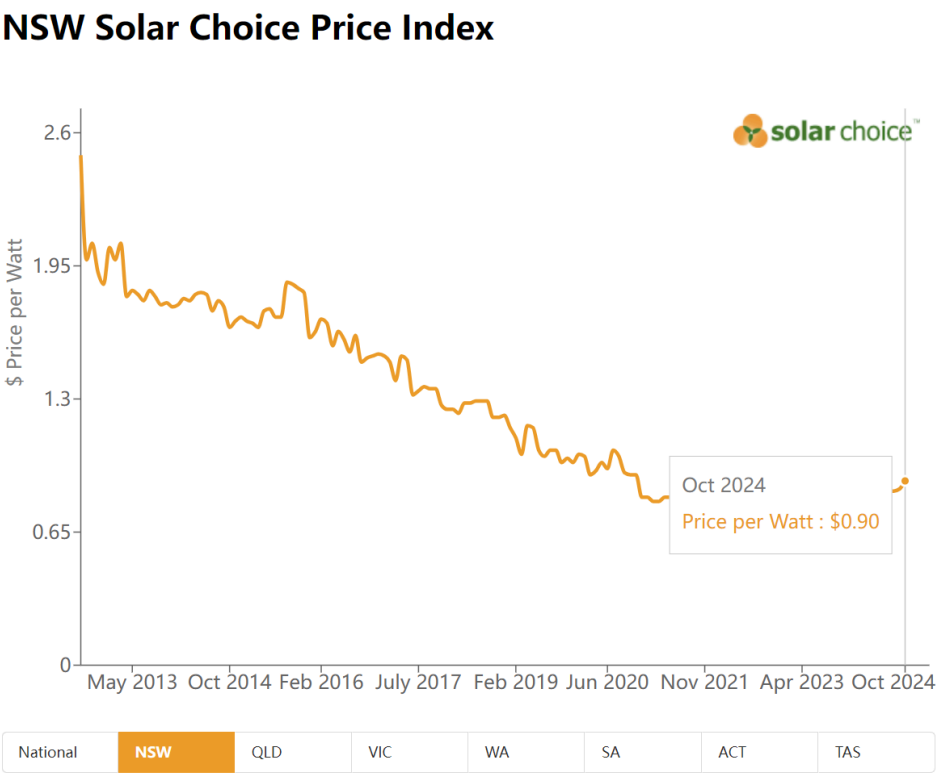



*Figure 5.1 NSW Solar Choice Price Index*

On the Solar Choice website, a 1MW solar farm in NSW costs about A$0.9 million in 2024, which is about 25% less than the expected total project cost. But we still use a rate of A$1,200 per kilowatt because it is a conservative estimate and in line with expectations.

Based on this estimate, a 20MW solar farm will cost about A$24 million.Considering the need to build a monitoring room and reserve space for future expansion to a 30MW solar farm, an additional A$1 million is expected to be required.

Therefore, a budget of A$25 million for the initial project is reasonable.

### 5.2.3 Bottom-Up Approach

To perform a bottom-up estimation we first subdivided the WBS into an activities list.

We then estimated the expected hours for each task. We used reasonable wage rates from the Rawlinsons Construction Cost Guide and estimated material costs based on available online information and market quotes.In our bottom-up approach, the estimated wages for workers at different levels are as follows:

| Level of | Labour | Salary per Hour |
|---|---|---|

| Grade | | (AUD) |
|---|---|---|
| 1 | General Worker | $40.1 |
| 2 | Advanced Machine Operator | $45.5 |
| 3 | Purchasing/Logistics Specialist/Clerical Staff | $48.7 |
| 4 | Project Team/Engineers | $55.0 |
| 5 | Project Manager/Senior Engineer | $65.1 |
| 6 | Senior technical expert | $77.5 |

*Table 5.1 Different Grades Standard Labour Rates*

Finally, we analyze it together with the Activity List in Section 4.2 to determine if there is room for improvement or if we are within the right budget.

## 5.2.4 Budget Table

### 5.2.4.1 Old Budget Table

| WBS | Working | | Activity | | Labour Grade | Duration(hours) | Duration(days) | Hourly Salary | Number | Salary Cost | Labour Cost | Material Cost | Total Cost |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Code | Description | Code | Description | | | | | | | | | |
| 1.1 Project System Design | 1.1.1 | Site Survey and Study | 1.1.1.1 | Preliminary site inspection | 4 | 160 | 20 | 55 | 4 | 35200 | 166128 | 20000 | 186128 |
| | | | 1.1.1.2 | Investigation and research execution | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | | | 1.1.1.3 | Data collection and analysis | 4 | 160 | 20 | 55 | 4 | 35200 | | | |
| | | | 1.1.1.4 | Stage meeting | 5 | 160 | 20 | 65.1 | 8 | 83328 | | | |
| | 1.1.2 | Geotechnical Report | 1.1.2.1 | Preparation before geological survey | 5 | 160 | 20 | 65.1 | 4 | 41664 | 154784 | 30000 | 184784 |
| | | | 1.1.2.2 | Site drilling and soil sampling | 2 | 120 | 15 | 45.5 | 12 | 65520 | | | |
| | | | 1.1.2.3 | Sample testing and analysis | 4 | 160 | 20 | 55 | 4 | 35200 | | | |
| | | | 1.1.2.4 | Data collection and collation | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | 1.1.3 | PV Cell Design | 1.1.3.1 | Requirements and parameter determination | 6 | 40 | 5 | 77.5 | 4 | 12400 | 129552 | 100000 | 229552 |
| | | | 1.1.3.2 | Preliminary design and simulation | 4 | 160 | 20 | 55 | 8 | 70400 | | | |
| | | | 1.1.3.3 | Component selection and procurement | 3 | 120 | 15 | 48.7 | 8 | 46752 | | | |
| | 1.1.4 | Electrical Plant Design | 1.1.4.1 | Cabling layout and Switchyard design | 4 | 160 | 20 | 55 | 8 | 70400 | 257360 | 100000 | 357360 |
| | | | 1.1.4.2 | Inverters arrangement design | 4 | 160 | 20 | 55 | 8 | 70400 | | | |
| | | | 1.1.4.3 | Connection to grid design | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | | | 1.1.4.4 | Monitoring Station Design | 5 | 160 | 20 | 65.1 | 5 | 52080 | | | |
| | | | 1.1.4.5 | Electrical Plant Design report | 5 | 160 | 20 | 65.1 | 5 | 52080 | | | |
| | 1.1.5 | Control & Instrumentation Specification | 1.1.5.1 | Demand analysis | 6 | 40 | 5 | 77.5 | 4 | 12400 | 89264 | 50000 | 139264 |
| | | | 1.1.5.2 | Draft specification preparation | 4 | 160 | 20 | 55 | 4 | 35200 | | | |
| | | | 1.1.5.3 | Technical review | 5 | 160 | 20 | 65.1 | 4 | 41664 | | | |
| 1.2 Mechanical Plant | 1.2.1 | PV Modules | 1.2.1.1 | Purchasing | 3 | 480 | 60 | 48.7 | 20 | 467520 | 5975520 | 645000 | 6620520 |
| | | | 1.2.1.2 | Manufacturing | 2 | 720 | 90 | 45.5 | 80 | 2620800 | | | |
| | | | 1.2.1.3 | Transportation and Installation | 1 | 720 | 90 | 40.1 | 100 | 2887200 | | | |
| | 1.2.2 | Steel Support Racks | 1.2.2.1 | Purchasing | 3 | 480 | 60 | 48.7 | 20 | 467520 | 5975520 | 300000 | 6275520 |
| | | | 1.2.2.2 | Manufacturing | 2 | 720 | 90 | 45.5 | 80 | 2620800 | | | |
| | | | 1.2.2.3 | Transportation and Installation | 1 | 720 | 90 | 40.1 | 100 | 2887200 | | | |
| 1.3 Electrical Infrastructure | 1.3.1 | Switchyard | 1.3.1.1 | Switchyard Purchasing | 3 | 160 | 20 | 48.7 | 10 | 77920 | 462880 | 1240000 | 1702880 |
| | | | 1.3.1.2 | Switchyard Transportation and Installation | 1 | 240 | 30 | 40.1 | 40 | 384960 | | | |
| | 1.3.2 | Transformers & Inverters | 1.3.2.1 | Transformers & Inverters Purchasing | 3 | 160 | 20 | 48.7 | 10 | 77920 | 462880 | 900000 | 1362880 |
| | | | 1.3.2.2 | Transformers & Inverters Transportation and Installation | 1 | 240 | 30 | 40.1 | 40 | 384960 | | | |
| | 1.3.3 | Cable Conduit System | 1.3.3.1 | Material Purchasing | 3 | 160 | 20 | 48.7 | 10 | 77920 | 462880 | 650000 | 1112880 |
| | | | 1.3.3.2 | Material Transportation and Installation | 1 | 240 | 30 | 40.1 | 40 | 384960 | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.4 Civil work | 1.4.1 | Earthworks | 1.4.1.1 | Land clearing | 1 | 80 | 10 | 40.1 | 30 | 96240 | 131440 | 150000 | 281440 |
| | | | 1.4.1.2 | Zone division | 4 | 80 | 10 | 55 | 8 | 35200 | | | |
| | 1.4.2 | PV Module Foundations | 1.4.2.1 | Steel Support Racks Installation | 1 | 80 | 10 | 40.1 | 20 | 64160 | 136960 | 100000 | 236960 |
| | | | 1.4.2.2 | PV Modules Installation | 2 | 80 | 10 | 45.5 | 20 | 72800 | | | |
| | 1.4.3 | Switchyard Foundations | 1.4.3.1 | Installation | 2 | 80 | 10 | 45.5 | 30 | 109200 | 109200 | 100000 | 209200 |
| | 1.4.4 | Reticulation Cabling trenches | 1.4.4.1 | Construction | 1 | 96 | 12 | 40.1 | 40 | 153984 | 153984 | 100000 | 253984 |
| | 1.4.5 | Roads | 1.4.5.1 | Materials Purchasing | 3 | 64 | 8 | 48.7 | 10 | 31168 | 200208 | 100000 | 300208 |
| | | | 1.4.5.2 | Materials Transportation | 1 | 80 | 10 | 40.1 | 30 | 96240 | | | |
| | | | 1.4.5.3 | Materials Installation | 2 | 80 | 10 | 45.5 | 20 | 72800 | | | |
| | 1.4.6 | Drainage | 1.4.6.1 | Drainage system design | 4 | 40 | 5 | 55 | 8 | 17600 | 188768 | 130000 | 318768 |
| | | | 1.4.6.2 | Materials Purchasing | 3 | 64 | 8 | 48.7 | 10 | 31168 | | | |
| | | | 1.4.6.3 | Materials Transportation | 1 | 80 | 10 | 40.1 | 20 | 64160 | | | |
| | | | 1.4.6.4 | Materials Installation | 2 | 64 | 8 | 45.5 | 20 | 58240 | | | |
| | | | 1.4.6.5 | System Testing | 4 | 40 | 5 | 55 | 8 | 17600 | | | |
| | 1.4.7 | Site Buildings | 1.4.7.1 | Materials Purchasing | 3 | 64 | 8 | 48.7 | 10 | 31168 | 234064 | 150000 | 384064 |
| | | | 1.4.7.2 | Materials Transportation | 1 | 80 | 10 | 40.1 | 30 | 96240 | | | |
| | | | 1.4.7.3 | Construction | 1 | 80 | 10 | 40.1 | 30 | 96240 | | | |
| | | | 1.4.7.4 | Completion Acceptance | 5 | 40 | 5 | 65.1 | 4 | 10416 | | | |
| 1.5 Monitoring & Control | 1.5.1 | Instruments | 1.5.1.1 | Purchasing | 3 | 64 | 8 | 48.7 | 10 | 31168 | 121488 | 300000 | 421488 |
| | | | 1.5.1.2 | Transportation | 1 | 40 | 5 | 40.1 | 20 | 32080 | | | |
| | | | 1.5.1.3 | Installation | 2 | 64 | 8 | 45.5 | 20 | 58240 | | | |
| | 1.5.2 | Site Controls | 1.5.2.1 | Control System Installation | 4 | 64 | 8 | 55 | 10 | 35200 | 35200 | 200000 | 235200 |
| | 1.5.3 | Telemetry | 1.5.3.1 | Software Design | 5 | 80 | 10 | 65.1 | 10 | 52080 | 67580 | 640000 | 707580 |
| | | | 1.5.3.2 | System Configuration | 6 | 40 | 5 | 77.5 | 5 | 15500 | | | |
| | 1.5.4 | UNSW Campus Control Room | 1.5.4.1 | Select a suitable room | 4 | 64 | 8 | 55 | 10 | 35200 | 35200 | 800000 | 835200 |
| 1.6 Testing and Commissioning | 1.6.1 | PerformanceTesting | 1.6.1.1 | Performance Testing | 6 | 720 | 90 | 77.5 | 2 | 111600 | 174096 | 100000 | 274096 |
| | | | 1.6.1.2 | Reporting | 5 | 240 | 30 | 65.1 | 4 | 62496 | | | |
| | 1.6.2 | Security Inspections | 1.6.2.1 | Security testing | 6 | 720 | 90 | 77.5 | 2 | 111600 | 174096 | 80000 | 254096 |
| | | | 1.6.2.2 | Reporting | 5 | 240 | 30 | 65.1 | 4 | 62496 | | | |
| | 1.6.3 | Operation Test Report | 1.6.3.1 | Operation Testing | 6 | 720 | 90 | 77.5 | 2 | 111600 | 174096 | 90000 | 264096 |
| | | | 1.6.3.2 | Reporting | 5 | 240 | 30 | 65.1 | 4 | 62496 | | | |
| | 1.6.4 | Stability Test report | 1.6.4.1 | Stability Testing | 6 | 720 | 90 | 77.5 | 2 | 111600 | 174096 | 100000 | 274096 |
| | | | 1.6.4.2 | Reporting | 5 | 240 | 30 | 65.1 | 4 | 62496 | | | |
| 1.7 Files & Documentation | 1.7.1 | Monitoring Training Documents | 1.7.1.1 | Develop documentation | 6 | 200 | 25 | 77.5 | 2 | 31000 | 31000 | 25000 | 56000 |
| | 1.7.2 | Equipment Handbooks | 1.7.2.1 | Develop documentation | 6 | 200 | 25 | 77.5 | 2 | 31000 | 31000 | 20000 | 51000 |
| | 1.7.3 | Surveillance Training Files | 1.7.3.1 | Develop documentation | 6 | 200 | 25 | 77.5 | 2 | 31000 | 31000 | 35000 | 66000 |
| | 1.7.4 | User Guides | 1.7.4.1 | Develop documentation | 6 | 120 | 15 | 77.5 | 2 | 18600 | 18600 | 15000 | 33600 |

## 5.2.4.1 New Budget Table

| WBS | Working | | Activity | | Labour Grade | Duration(hours) | Duration(days) | Hourly Salary | Number | Salary Cost | Labour Cost | Material Cost | Total Cost |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Code | Description | Code | Description | | | | | | | | | |
| 1.1 Project System Design | 1.1.1 | Site Survey and Study | 1.1.1.1 | Preliminary site inspection | 4 | 96 | 12 | 55 | 8 | 42240 | 171875.2 | 20000 | 191875.2 |
| | | | 1.1.1.2 | Investigation and research execution | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | | | 1.1.1.3 | Data collection and analysis | 4 | 96 | 12 | 55 | 8 | 42240 | | | |
| | | | 1.1.1.4 | Stage meeting | 5 | 96 | 12 | 65.1 | 12 | 74995.2 | | | |
| | 1.1.2 | Geotechnical Report | 1.1.2.1 | Preparation before geological survey | 5 | 96 | 12 | 65.1 | 8 | 49996.8 | 157052.8 | 30000 | 187052.8 |
| | | | 1.1.2.2 | Site drilling and soil sampling | 2 | 72 | 9 | 45.5 | 16 | 52416 | | | |
| | | | 1.1.2.3 | Sample testing and analysis | 4 | 96 | 12 | 55 | 8 | 42240 | | | |
| | | | 1.1.2.4 | Data collection and collation | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | 1.1.3 | PV Cell Design | 1.1.3.1 | Requirements and parameter determination | 6 | 40 | 5 | 77.5 | 4 | 12400 | 122512 | 100000 | 222512 |
| | | | 1.1.3.2 | Preliminary design and simulation | 4 | 96 | 12 | 55 | 12 | 63360 | | | |
| | | | 1.1.3.3 | Component selection and procurement | 3 | 80 | 10 | 48.7 | 12 | 46752 | | | |
| | 1.1.4 | Electrical Plant Design | 1.1.4.1 | Cabling layout and Switchyard design | 4 | 96 | 12 | 55 | 12 | 63360 | 264112 | 100000 | 364112 |
| | | | 1.1.4.2 | Inverters arrangement design | 4 | 96 | 12 | 55 | 12 | 63360 | | | |
| | | | 1.1.4.3 | Connection to grid design | 6 | 40 | 5 | 77.5 | 4 | 12400 | | | |
| | | | 1.1.4.4 | Monitoring Station Design | 5 | 96 | 12 | 65.1 | 10 | 62496 | | | |
| | | | 1.1.4.5 | Electrical Plant Design report | 5 | 96 | 12 | 65.1 | 10 | 62496 | | | |
| | 1.1.5 | Control & Instrumentation Specification | 1.1.5.1 | Demand analysis | 6 | 40 | 5 | 77.5 | 4 | 12400 | 104636.8 | 50000 | 154636.8 |
| | | | 1.1.5.2 | Draft specification preparation | 4 | 96 | 12 | 55 | 8 | 42240 | | | |
| | | | 1.1.5.3 | Technical review | 5 | 96 | 12 | 65.1 | 8 | 49996.8 | | | |
| | 1.1.6 | Additional Control Instrumentation Design | 1.1.6.1 | Design Additional Control Instrumentation | 5 | 96 | 12 | 65.1 | 8 | 49996.8 | 49996.8 | 20000 | 69996.8 |
| 1.2 Mechanical Plant | 1.2.1 | PV Modules | 1.2.1.1 | Purchasing | 3 | 320 | 40 | 48.7 | 40 | 623360 | 6131360 | 645000 | 6776360 |
| | | | 1.2.1.2 | Manufacturing | 2 | 480 | 60 | 45.5 | 120 | 2620800 | | | |
| | | | 1.2.1.3 | Transportation and Installation | 1 | 480 | 60 | 40.1 | 150 | 2887200 | | | |
| | 1.2.2 | Steel Support Racks | 1.2.2.1 | Purchasing | 3 | 320 | 40 | 48.7 | 40 | 623360 | 6131360 | 300000 | 6431360 |
| | | | 1.2.2.2 | Manufacturing | 2 | 480 | 60 | 45.5 | 120 | 2620800 | | | |
| | | | 1.2.2.3 | Transportation and Installation | 1 | 480 | 60 | 40.1 | 150 | 2887200 | | | |
| 1.3 Electrical Infrastructure | 1.3.1 | Switchyard | 1.3.1.1 | Switchyard Purchasing | 3 | 96 | 12 | 48.7 | 18 | 84153.6 | 488361.6 | 1240000 | 1728361.6 |
| | | | 1.3.1.2 | Switchyard Transportation and Installation | 1 | 144 | 18 | 40.1 | 70 | 404208 | | | |
| | 1.3.2 | Transformers & Inverters | 1.3.2.1 | Transformers & Inverters Purchasing | 3 | 96 | 12 | 48.7 | 18 | 84153.6 | 488361.6 | 900000 | 1388361.6 |
| | | | 1.3.2.2 | Transformers & Inverters Transportation and Installation | 1 | 144 | 18 | 40.1 | 70 | 404208 | | | |
| | 1.3.3 | Cable Conduit System | 1.3.3.1 | Material Purchasing | 3 | 96 | 12 | 48.7 | 18 | 84153.6 | 488361.6 | 650000 | 1138361.6 |
| | | | 1.3.3.2 | Material Transportation and Installation | 1 | 144 | 18 | 40.1 | 70 | 404208 | | | |
| 1.4 Civil work | 1.4.1 | Earthworks | 1.4.1.1 | Land clearing | 1 | 48 | 6 | 40.1 | 50 | 96240 | 127920 | 150000 | 277920 |
| | | | 1.4.1.2 | Zone division | 4 | 48 | 6 | 55 | 12 | 31680 | | | |
| | 1.4.2 | PV Module Foundations | 1.4.2.1 | Steel Support Racks Installation | 1 | 48 | 6 | 40.1 | 30 | 57744 | 123264 | 100000 | 223264 |
| | | | 1.4.2.2 | PV Modules Installation | 2 | 48 | 6 | 45.5 | 30 | 65520 | | | |
| | 1.4.3 | Switchyard Foundations | 1.4.3.1 | Installation | 2 | 48 | 6 | 45.5 | 50 | 109200 | 109200 | 100000 | 209200 |
| | 1.4.4 | Reticulation Cabling trenches | 1.4.4.1 | Construction | 1 | 72 | 9 | 40.1 | 70 | 202104 | 202104 | 100000 | 302104 |
| | 1.4.5 | Roads | 1.4.5.1 | Materials Purchasing | 3 | 48 | 6 | 48.7 | 18 | 42076.8 | 203836.8 | 100000 | 303836.8 |
| | | | 1.4.5.2 | Materials Transportation | 1 | 48 | 6 | 40.1 | 50 | 96240 | | | |
| | | | 1.4.5.3 | Materials Installation | 2 | 48 | 6 | 45.5 | 30 | 65520 | | | |
| | 1.4.6 | Drainage | 1.4.6.1 | Drainage system design | 4 | 32 | 4 | 55 | 12 | 21120 | 212860.8 | 130000 | 342860.8 |
| | | | 1.4.6.2 | Materials Purchasing | 3 | 48 | 6 | 48.7 | 18 | 42076.8 | | | |
| | | | 1.4.6.3 | Materials Transportation | 1 | 48 | 6 | 40.1 | 30 | 57744 | | | |
| | | | 1.4.6.4 | Materials Installation | 2 | 48 | 6 | 45.5 | 30 | 65520 | | | |
| | | | 1.4.6.5 | System Testing | 4 | 32 | 4 | 55 | 15 | 26400 | | | |
| | 1.4.7 | Site Buildings | 1.4.7.1 | Materials Purchasing | 3 | 48 | 6 | 48.7 | 15 | 35064 | 240043.2 | 150000 | 390043.2 |
| | | | 1.4.7.2 | Materials Transportation | 1 | 48 | 6 | 40.1 | 50 | 96240 | | | |
| | | | 1.4.7.3 | Construction | 1 | 48 | 6 | 40.1 | 50 | 96240 | | | |
| | | | 1.4.7.4 | Completion Acceptance | 5 | 32 | 4 | 65.1 | 6 | 12499.2 | | | |
| 1.5 Monitoring & Control | 1.5.1 | Instruments | 1.5.1.1 | Purchasing | 3 | 64 | 8 | 48.7 | 18 | 56102.4 | 281902.4 | 600000 | 881902.4 |
| | | | 1.5.1.2 | Transportation | 1 | 40 | 5 | 40.1 | 50 | 80200 | | | |
| | | | 1.5.1.3 | Installation | 2 | 64 | 8 | 45.5 | 50 | 145600 | | | |
| | 1.5.2 | Additional Control Instrumentation | 1.5.2.1 | Purchasing | 3 | 80 | 10 | 48.7 | 18 | 70128 | 663928 | 800000 | 1463928 |
| | | | 1.5.2.2 | Transportation | 1 | 160 | 20 | 40.1 | 50 | 320800 | | | |
| | | | 1.5.2.3 | Installation | 2 | 120 | 15 | 45.5 | 50 | 273000 | | | |
| | 1.5.3 | Site Controls | 1.5.3.1 | Control System Installation | 4 | 64 | 8 | 55 | 50 | 176000 | 176000 | 200000 | 376000 |
| | 1.5.4 | Telemetry | 1.5.4.1 | Software Design | 5 | 64 | 8 | 65.1 | 20 | 83328 | 95728 | 640000 | 735728 |
| | | | 1.5.4.2 | System Configuration | 6 | 32 | 4 | 77.5 | 5 | 12400 | | | |
| | 1.5.5 | UNSW Campus Control Room | 1.5.5.1 | Select a suitable room | 4 | 48 | 6 | 55 | 12 | 31680 | 31680 | 800000 | 831680 |
| 1.6 Testing and Commissioning | 1.6.1 | PerformanceTesting | 1.6.1.1 | Performance Testing | 6 | 480 | 60 | 77.5 | 4 | 148800 | 232128 | 100000 | 332128 |
| | | | 1.6.1.2 | Reporting | 5 | 160 | 20 | 65.1 | 8 | 83328 | | | |
| | 1.6.2 | Security Inspections | 1.6.2.1 | Security testing | 6 | 480 | 60 | 77.5 | 4 | 148800 | 232128 | 80000 | 312128 |
| | | | 1.6.2.2 | Reporting | 5 | 160 | 20 | 65.1 | 8 | 83328 | | | |
| | 1.6.3 | Operation Test Report | 1.6.3.1 | Operation Testing | 6 | 480 | 60 | 77.5 | 4 | 148800 | 232128 | 90000 | 322128 |
| | | | 1.6.3.2 | Reporting | 5 | 160 | 20 | 65.1 | 8 | 83328 | | | |
| | 1.6.4 | Stability Test report | 1.6.4.1 | Stability Testing | 6 | 480 | 60 | 77.5 | 4 | 148800 | 232128 | 100000 | 332128 |
| | | | 1.6.4.2 | Reporting | 5 | 160 | 20 | 65.1 | 8 | 83328 | | | |
| 1.7 Files & Documentation | 1.7.1 | Monitoring Training Documents | 1.7.1.1 | Develop documentation | 6 | 150 | 18.75 | 77.5 | 4 | 46500 | 46500 | 25000 | 71500 |
| | 1.7.2 | Equipment Handbooks | 1.7.2.1 | Develop documentation | 6 | 150 | 18.75 | 77.5 | 4 | 46500 | 46500 | 20000 | 66500 |
| | 1.7.3 | Surveillance Training Files | 1.7.3.1 | Develop documentation | 6 | 150 | 18.75 | 77.5 | 4 | 46500 | 46500 | 35000 | 81500 |
| | 1.7.4 | User Guides | 1.7.4.1 | Develop documentation | 6 | 80 | 10 | 77.5 | 4 | 24800 | 24800 | 15000 | 39800 |

## 5.2.5 Conclusion

### 5.2.5.1 Old Budget Summary

| Control Account | Direct Labour Cost(AUD) | Material Cost(AUD) | Total Work Package Cost(AUD) | Contingency(AUD) | Total Cost |
|---|---|---|---|---|---|
| 1.1 Project System Design | $797,088 | $300,000 | $1,097,088 | $40,000 | $1,137,088 |
| 1.2 Mechanical Plant | $11,951,040 | $945,000 | $12,896,040 | $680,000 | $13,576,040 |
| 1.3 Electrical Infrastructure | $1,388,640 | $2,790,000 | $4,178,640 | $400,000 | $4,578,640 |
| 1.4 Civil work | $1,154,624 | $830,000 | $1,984,624 | $100,000 | $2,084,624 |
| 1.5 Monitoring & Control | $259,468 | $1,940,000 | $2,199,468 | $60,000 | $2,259,468 |
| 1.6 Testing and Commissioning | $696,384 | $370,000 | $1,066,384 | $70,000 | $1,136,384 |
| 1.7 Files & Documentation | $111,600 | $95,000 | $206,600 | $20,000 | $226,600 |
| Total Cost | $16,358,844 | $7,270,000 | $23,628,844 | $1,370,000 | $24,998,844 |

*Old Table Budget Summary*

### 5.2.5.2 New Budget Summary

| Control Account | Direct Labour Cost (AUD) | Material Cost (AUD) | Total Work Package Cost (AUD) | Contingency (AUD) | Total Cost (AUD) |
|---|---|---|---|---|---|
| 1.1 Project System Design | $870, 185 | $320, 000 | $1, 190, 185 | $40, 000 | $1, 230, 185 |
| 1.2 Mechanical Plant | $12, 262, 720 | $945, 000 | $13, 207, 720 | $750, 000 | $13, 957, 720 |
| 1.3 Electrical Infrastructure | $1, 465, 084 | $2, 790, 000 | $4, 255, 084 | $500, 000 | $4, 755, 084 |
| 1.4 Civil Work | $1, 219, 228 | $830, 000 | $2, 049, 228 | $120, 000 | $2, 169, 228 |
| 1.5 Monitoring & Control | $1, 249, 238 | $3, 040, 000 | $4, 289, 238 | $72, 000 | $4, 361, 238 |
| 1.6 Testing and Commissioning | $928, 512 | $370, 000 | $1, 298, 512 | $84, 000 | $1, 382, 512 |
| 1.7 Files & Documentation | $164, 300 | $95, 000 | $259, 300 | $20, 000 | $279, 300 |
| Total Cost | $18, 159, 267 | $8, 390, 000 | $26, 549, 267 | $1, 586, 000 | $28, 135, 267 |

*New Table Budget Summary*

### 5.2.5.3 Contingency plan

Contingency plans at different stages are as follows:

1. Project System Design :
   Allocate extra time for re-testing and troubleshooting
   Tighten cost management practices
   Allocate rework budget and schedule buffer
2. Mechanical Plant :
   Stock critical materials if possible
   Keep spare panels for immediate replacement
   Hire additional surveying teams
   Continuous safety training and audits
3. Electrical Infrastructure:
   Work with legal team to expedite processes
   Have alternative suppliers ready
4. Civil work:
   Establish disaster insurance coverage
   Create alternative sourcing plans
   Establish emergency medical response on site
5. Monitoring & Control:
   Monitor weather forecasts, adjust schedules
   Have technical teams standby

Keep technical teams on call

6. Testing and Commissioning:
   Establish legal team to handle disputes
   Keep structural experts on call

5.2.5.4 Concusion

Based on the bottom-up approach, we estimated the budget for the entire project to be US$23.5 million (without considering any contingencies), which was within the client's budget requirement of A$25 million, taking into account the funds required to deal with any contingencies.

Due to the updated project plan requirements, the project needs to be completed within 11 months. In order to advance the progress faster, we need to increase the number of workers throughout the project, and increase contingency cost to deal with greater risk projects. Based on the above considerations, we have concluded that new project budget is approximately A$28.1 million.

## 5.3 Integrated Project Budget (Time Phased Budget)

| | | | 24q4 | | | 25q1 | | | 25q2 | | | 25q3 | | | 25q4 | | | 26q1 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| WBS | Activity | Duration | Oct | Nov | Dec | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Jan | Feb | Material Cost | Labour Cost | Budget |
| 1.1.1.1 | Preliminary site inspection | 20 | 40200 | | | | | | | | | | | | | | | | | | 35290 | |
| 1.1.1.2 | Investigation and research execution | 5 | 17400 | | | | | | | | | | | | | | | | | | 12400 | |
| 1.1.1.3 | Data collection and analysis | 20 | 40200 | | | | | | | | | | | | | | | | | 20000 | 35290 | 188128 |
| 1.1.1.4 | Stage meeting | 20 | 88328 | | | | | | | | | | | | | | | | | | 83328 | |
| 1.1.2.1 | Preparation before geological survey | 20 | | 49164 | | | | | | | | | | | | | | | | | 41694 | |
| 1.1.2.2 | Site drilling and soil sampling | 15 | | | 73020 | | | | | | | | | | | | | | | | 65520 | |
| 1.1.2.3 | Sample testing and analysis | 20 | | | 42700 | | | | | | | | | | | | | | | 30000 | 35290 | 184784 |
| 1.1.2.4 | Data collection and collation | 5 | | | | 19900 | | | | | | | | | | | | | | | 12490 | |
| 1.1.3.1 | Requirements and parameter determination | 5 | | | | 45733.33333 | | | | | | | | | | | | | | | 12490 | |
| 1.1.3.2 | Preliminary design and simulation | 20 | | | | 103733.3333 | | | | | | | | | | | | | | 100000 | 76490 | 229552 |
| 1.1.3.3 | Component selection and procurement | 15 | | | | 80085.33333 | | | | | | | | | | | | | | | 46752 | |
| 1.1.4.1 | Cabling layout and Switchyard design | 20 | | | | | 90400 | | | | | | | | | | | | | | 76490 | |
| 1.1.4.2 | Inverters arrangement design | 20 | | | | | 90400 | | | | | | | | | | | | | | 76490 | |
| 1.1.4.3 | Connection to grid design | 5 | | | | | 32400 | | | | | | | | | | | | | 100000 | 12490 | 357360 |
| 1.1.4.4 | Monitoring Station Design | 20 | | | | | | 72080 | | | | | | | | | | | | | 52080 | |
| 1.1.4.5 | Electrical Plant Design report | 20 | | | | | | 72080 | | | | | | | | | | | | | 52080 | |
| 1.1.5.1 | Demand analysis | 5 | | | | | | 29066.66667 | | | | | | | | | | | | | 12490 | |
| 1.1.5.2 | Draft specification preparation | 20 | | | | | | 51866.66667 | | | | | | | | | | | | 50000 | 35290 | 139264 |
| 1.1.5.3 | Technical review | 20 | | | | | | 58330.66667 | | | | | | | | | | | | | 41694 | |
| 1.2.1.1 | Purchasing | 60 | | | | | | | 682520 | | | | | | | | | | | | 467520 | |
| 1.2.1.2 | Manufacturing | 90 | | | | | | | 2835800 | | | | | | | | | | | 645000 | 2620800 | 6620520 |
| 1.2.1.3 | Transportation and Installation | 90 | | | | | | | 3102200 | | | | | | | | | | | | 2887200 | |
| 1.2.2.1 | Purchasing | 60 | | | | | | | | 567520 | | | | | | | | | | | 467520 | |
| 1.2.2.2 | Manufacturing | 90 | | | | | | | | 2720800 | | | | | | | | | | 300000 | 2620800 | 6275520 |
| 1.2.2.3 | Transportation and Installation | 90 | | | | | | | | 2987200 | | | | | | | | | | | 2887200 | |
| 1.3.1.1 | Switchyard Purchasing | 20 | | | | | | | | 697920 | | | | | | | | | | 1240000 | 77920 | 1702880 |
| 1.3.1.2 | Switchyard Transportation and Installation | 30 | | | | | | | | 1004960 | | | | | | | | | | | 384960 | |
| 1.3.2.1 | Transformers & Inverters Purchasing | 20 | | | | | | | | | 527920 | | | | | | | | | 900000 | 77920 | 1362880 |
| 1.3.2.2 | Transformers & Inverters Transportation and Installation | 30 | | | | | | | | | 834960 | | | | | | | | | | 384960 | |
| 1.3.3.1 | Material Purchasing | 20 | | | | | | | | | 402920 | | | | | | | | | 650000 | 77920 | 1112880 |
| 1.3.3.2 | Material Transportation and Installation | 30 | | | | | | | | | | 709960 | | | | | | | | | 384960 | |
| 1.4.1.1 | Land clearing | 10 | | | | | | | | | | 171240 | | | | | | | | 150000 | 96240 | 281440 |
| 1.4.1.2 | Zone division | 10 | | | | | | | | | | 110200 | | | | | | | | | 35290 | |
| 1.4.2.1 | Steel Support Racks Installation | 10 | | | | | | | | | | 114160 | | | | | | | | 100000 | 64160 | 236960 |
| 1.4.2.2 | PV Modules Installation | 10 | | | | | | | | | | 122800 | | | | | | | | | 72800 | |
| 1.4.3.1 | Installation | 10 | | | | | | | | | | | 209200 | | | | | | | 100000 | 109200 | 209200 |
| 1.4.4.1 | Construction | 12 | | | | | | | | | | | 259864 | | | | | | | 100000 | 153864 | 253864 |
| 1.4.5.1 | Materials Purchasing | 8 | | | | | | | | | | | 64501.33333 | | | | | | | | 31168 | |
| 1.4.5.2 | Materials Transportation | 10 | | | | | | | | | | | 129573.3333 | | | | | | | 100000 | 96240 | 309208 |
| 1.4.5.3 | Materials Installation | 10 | | | | | | | | | | | 106133.3333 | | | | | | | | 72800 | |
| 1.4.6.1 | Drainage system design | 5 | | | | | | | | | | | | 43600 | | | | | | | 17600 | |
| 1.4.6.2 | Materials Purchasing | 8 | | | | | | | | | | | | 57168 | | | | | | | 31168 | |
| 1.4.6.3 | Materials Transportation | 10 | | | | | | | | | | | | 90160 | | | | | | 130000 | 64160 | 318768 |
| 1.4.6.4 | Materials Installation | 8 | | | | | | | | | | | | 84240 | | | | | | | 58240 | |
| 1.4.6.5 | System Testing | 5 | | | | | | | | | | | | 43600 | | | | | | | 17600 | |
| 1.4.7.1 | Materials Purchasing | 8 | | | | | | | | | | | | 68668 | | | | | | | 31168 | |
| 1.4.7.2 | Materials Transportation | 10 | | | | | | | | | | | | 133740 | | | | | | 150000 | 96240 | 364064 |
| 1.4.7.3 | Construction | 10 | | | | | | | | | | | | 133740 | | | | | | | 96240 | |
| 1.4.7.4 | Completion Acceptance | 5 | | | | | | | | | | | | 47916 | | | | | | | 18416 | |
| 1.5.1.1 | Purchasing | 8 | | | | | | | | | | | | | 131168 | | | | | | 31168 | |
| 1.5.1.2 | Transportation | 5 | | | | | | | | | | | | | 132080 | | | | | 300000 | 32080 | 421488 |
| 1.5.1.3 | Installation | 8 | | | | | | | | | | | | | 158240 | | | | | | 58240 | |
| 1.5.2.1 | Control System Installation | 8 | | | | | | | | | | | | | 235200 | | | | | 200000 | 35290 | 235200 |
| 1.5.3.1 | Software Design | 10 | | | | | | | | | | | | | 372080 | | | | | 640000 | 52080 | 707580 |
| 1.5.3.2 | System Configuration | 5 | | | | | | | | | | | | | 335500 | | | | | | 15580 | |
| 1.5.4.1 | Select a suitable room | 8 | | | | | | | | | | | | | 835200 | | | | | 800000 | 35290 | 835200 |
| 1.6.1.1 | Performance Testing | 90 | | | | | | | | | | | | | | | 161600 | | | | 111600 | |
| 1.6.1.2 | Reporting | 30 | | | | | | | | | | | | | | | 112496 | | | 100000 | 62496 | 274096 |
| 1.6.2.1 | Security testing | 90 | | | | | | | | | | | | | | | 151600 | | | | 111600 | |
| 1.6.2.2 | Reporting | 30 | | | | | | | | | | | | | | | 102496 | | | 80000 | 62496 | 254096 |
| 1.6.3.1 | Operation Testing | 90 | | | | | | | | | | | | | | | 156600 | | | | 111600 | |
| 1.6.3.2 | Reporting | 30 | | | | | | | | | | | | | | | 107496 | | | 90000 | 62496 | 264096 |
| 1.6.4.1 | Stability Testing | 90 | | | | | | | | | | | | | | | 161600 | | | | 111600 | |
| 1.6.4.2 | Reporting | 30 | | | | | | | | | | | | | | | 112496 | | | 100000 | 62496 | 274096 |
| 1.7.1.1 | Develop documentation | 25 | | | | | | | | | | | | | | | | | 56000 | 25000 | 31000 | 56000 |
| 1.7.2.1 | Develop documentation | 25 | | | | | | | | | | | | | | | | | 51000 | 20000 | 31000 | 51000 |
| 1.7.3.1 | Develop documentation | 25 | | | | | | | | | | | | | | | | | 66000 | 35000 | 31000 | 66000 |
| 1.7.4.1 | Develop documentation | 15 | | | | | | | | | | | | | | | | | 33600 | 15000 | 18600 | 33600 |
| | total cost | 1706 | 186128 | 164884 | 249452 | 357360 | 139264 | 6620520 | 6275520 | 1702880 | 1765800 | 1228360 | 763392 | 702832 | 2199468 | | 1066384 | | 206600 | 7270000 | 16358844 | 23628844 |

new time phased budget

# 6 Risk Management

## 6.1 Risk Breakdown Structure

### 6.1.1 Risk Identification

The initial stage of the Risk Breakdown Strategy involves identifying risks. This process includes collaborating with project stakeholders, seeking input from subject matter experts, and analyzing historical data to pinpoint potential risks that could affect the project. Risks are then classified according to their characteristics and recorded in the Risk Register.

### 6.1.2 Risk Categorisation

The identified risks have been organized into subgroups that cover various aspects, including technical, budget/scheduling, management, commercial, logistics, and external factors. This organization facilitates a systematic approach to addressing different types of risks and evaluating any emerging trends.

| Risk ID | Description | Risk Category | Owner |
|---|---|---|---|
| 1 | Critical materials supply chain delays. | Logistics | Project Manager |
| 2 | Permit and compliance issues | Management | Peoject Manager |
| 3 | Failed tests at commision | Technical | Project Manager |
| 4 | Construction material delays | Logistical | Project Manager |
| 5 | Worker injured | Technical | Project Manager |
| 6 | Electrical malfunctions | Technical | Project Manager |
| 7 | Unexpected weather conditions | External | Project Manager |
| 8 | Compatibility issues with PV modules | Technical | Project Manager |
| 9 | Solar panel failures | Technical | Project Manager |
| 10 | Natural desaster | External | Project Manager |
| 11 | Budget changes due to unforeseen expenses | Commercial | Project Manager |
| 12 | Civil rework due to design issues | Technical | Project Manager |
| 13 | Site survey delays | Management | Project Manager |
| 14 | Safety incidents | External | Project Manager |
| 15 | Community property disputes | External | Project Manager, UNSW |
| 16 | Structural issues with support racks. | Technical | Project Manager |

| Risk ID | Description | Risk Category | Owner |
|---|---|---|---|
| 1 | Critical materials supply chain delays. | Logistics | Project Manager |
| 2 | Permit and compliance issues | Management | Peoject Manager |
| 17 | Monitoring system failures | Technical | Project Manager |

# 6.2 Risk Assessment

## 6.2.1 Risk Assessment

Risks are evaluated through both qualitative and quantitative approaches. During the initial phase of the project, the focus has mainly been on qualitative assessments. However, as more data emerges and the design phase progresses, it is anticipated that the Risk Register will be revised to incorporate updated quantitative assessments.

## 6.2.2 Risk Severity Matrix

| | | Severity | | | | |
|---|---|---|---|---|---|---|
| | | Insignificant (1) | Minor (2) | Moderate (3) | Major (4) | Critical (5) |
| Likelihood | Very High (5) | L (5) | M (10) | H (15) | E (20) | E (25) |
| | High (4) | L (4) | M (8) | H (12) | E (16) | E (20) |
| | Medium (3) | L (3) | M (6) | M (9) | H (12) | H (15) |
| | Low (2) | L (2) | L (4) | M (6) | M (8) | M (10) |
| | Very Low (1) | L (1) | L (2) | L (3) | L (4) | L (5) |

E (16-25): Extremely high risk, immediate action is required
H (11-15): High risk, additional research and measurements required
M (6-10): Moderate risk, mitigation required
L (0-5): Low risk, minor control and monitoring required

# 6.3 Risk Register (Response and Contingency Plan) (Change in Tab2)

## 6.3.1 Risk Register Table ~~(split to 6.1.3 and 6.3.3)~~ (More on Tab 2 !!!)

| Risk ID | Severity | Likelihood | Risk Assessment | Overall Severity | Response Strategy | Severiry (After) | Likelihood (After) |
|---|---|---|---|---|---|---|---|
| 1 | 5 | 4 | 20 | Extremely high | Mitigate | 5 | 2 |
| 2 | 4 | 3 | 12 | High | Mitigate | 4 | 2 |
| 3 | 3 | 3 | 9 | Moderate | Mitigate | 3 | 2 |
| 4 | 3 | 3 | 9 | Moderate | Mitigate | 3 | 2 |
| 5 | 4 | 3 | 12 | High | Avoid | 4 | 1 |
| 6 | 3 | 3 | 9 | Moderate | Mitigate | 2 | 2 |
| 7 | 3 | 3 | 9 | Moderate | Accept | 2 | 2 |
| 8 | 4 | 2 | 8 | Moderate | Mitigate | 4 | 1 |
| 9 | 3 | 2 | 6 | Moderate | Mitigate | 2 | 2 |
| 10 | 5 | 1 | 5 | Low | Accept | 4 | 1 |
| 11 | 4 | 3 | 12 | High | Mitigate | 3 | 1 |
| 12 | 4 | 2 | 8 | Moderate | Mitigate | 3 | 1 |
| 13 | 3 | 2 | 6 | Moderate | Mitigate | 3 | 1 |
| 14 | 3 | 2 | 6 | Moderate | Avoid | 3 | 1 |
| 15 | 3 | 3 | 9 | Moderate | Mitigate | 2 | 1 |
| 16 | 3 | 2 | 6 | Moderate | Mitigate | 2 | 1 |
| 17 | 3 | 2 | 6 | Moderate | Mitigate | 2 | 1 |

## 6.3.2 RIsk Prioritisation

Risk prioritisation involves ranking risks based on their overall severity to determine which risks need immediate attention and which can be monitored with less urgency. The prioritisation is done based on a combination of likelihood and severity to focus efforts on the most critical risks.

## 6.3.3 Contingency Plan

| Risk ID | Response Action | Contingency Plan |
|---|---|---|
| 1 | Source multiple suppliers | Stock critical materials if possible |
| 2 | Early engagement with regulatory bodies | Work with legal team to expedite processes |
| 3 | Run multiple simulations tests | Allocate extra time for re-testing and troubleshooting |

| | | |
|---|---|---|
| 4 | Pre-order construction materials with time to buffer | Create alternative sourcing plans |
| 5 | Safety training and protocols | - |
| 6 | Regular system checks and maintenance | Have technical teams standby |
| 7 | Design weather-resistant infrastructure | - |
| 8 | Test module compatibility | Have alternative suppliers ready |
| 9 | Test panels for durability | Keep spare panels for immediate replacement |
| 10 | Disaster recovery plan | - |
| 11 | Establish budget contingency fund | Tighten cost management practices |
| 12 | Conduct thorough design reviews | Allocate rework budget and schedule buffer |
| 13 | Begin site surveys early | Hire additional surveying teams |
| 14 | Increase site safety inspections | - |
| 15 | Early community engagement | Establish legal team to handle disputes |
| 16 | Conduct structural testing and quality assurance | Keep structural experts on call |
| 17 | Test system integration | Keep technical teams on call |

## 6.4 Overview of Risk Management (PMBOK Referenced Discussion)

We used the knowledge from section 11 of PMBOK to manage the project risk. The first step is to identify risks, we use our expert judgment and some techniques like brainstorming and hold meetings to create a list about potential risks as shown in section 6.2 in this report.

The next step is about the assessment of risks. Risks need to be assessed from qualitative and quantitative perspectives. According to the likelihood and severity of risks, we can make an overall assessment for every risk. Then we can rank the risks based on their overall severity.
Each risk need a response action and a contingency plan in case the likelihood or impact of the risk. A contingency plan allows for quick action if an incident does occur and will help to reduce the impact of the risk.

# 7. Communications Management Plan

## 7.1 Project Information System Table

| Reporting Document | Document Description | Document Type | When | How | From | To |
|---|---|---|---|---|---|---|
| Meeting Agenda | Agenda that will go through during meeting | Semi-Live Document | At least 24 hours before | Email | Project Manager | Relevant Teams |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | meeting | | | |
| Meeing Minutes | Record of meeting minutes and key point | Semi-Live Document | No later than 24 hours after meeting | Email | Project Manager | Relevant Teams |
| Action Item List | A TODO list contain tasks assigned to each team and their deadline | Live Document | Beginning and end of all milestones with weekly update | Email | Project Manager | Relevant Teams |
| Progress Report | Project progress report based on schedule and budget, contains achievements, challenges and future plans. | Live Document | Beginning and end of all milestones with weekly update | Email | Project Manager | Relevant Teams |
| Change Request | Requests to change project plans including project scope or requirements. | Semi-Live Document | Created when a change is required | Email | Project Manager | Relevant Team and Stakeholders |
| Quality Assurance Plan | Project quality standards for testing | Semi-Live Document | Developed during project planning and update when needed | Email | Project Manager | Relevant Teams and Stakeholders |
| Risk Response and Contingency Plan | Contains risk identification, risk assessments, risk response and contingency plans. | Semi-Live Document | Developed during project planning and update when needed | Email | Project Manager | Relevant Teams and Stakeholders |
| Training documents | Guidelines for each team training and support materials | Permanent Document | Developed during project planning and update when needed | Email | Project Manager | Relevant Teams |
| Environmental Impact Report | Report of the environmental impact caused by the project | Permanent Document | Prepared as required by the environmental regulatories. | Email | Project Manager | Relevant Teams and Stakeholders |

## 7.2 Project Communication Plan - changed

| Communication Type | Objective | Medium | Frequency | Audience | Owner | Deliverable Format |
|---|---|---|---|---|---|---|
| Start Up Meeting | Introduce project and stakeholders | Face to Face Presentation | Project initiation | Sponsor Project Team Stakeholders | Project Manager | Meeting Agenda Meeting Minutes |
| Sponsor Meeting | Explain project scope to sponsor in detail | Face to Face Online Meeting | Weekly | Project Sponsor | Project Manager | Meeting Agenda Meeting Minutes Project Scope |
| Project Updates Meeting | Update project progress and milestones | Face to Face Online Meetings Eamil | Weekly WIth beginning and end of milestone | Project Teams Project Sponsor Relevant Stakeholders | Project Manager | Meeting Agenda Meeting Minutes Progress Reports |
| Stakeholder Meeting | Engage with stakeholder including UNSW researchers, UNSW community and UNSW Facilities Management | Face to Face Online Meetings Email | Weekly or As needed | UNSW researchers UNSW community UNSW facilities management | Project Manager | Meeting Agenda Meeting Minutes |
| Engineering Teams | Coordinate decisions related to engineer design | Face to Face Online Meetings Email | Weekly | Engineering Team | Project Manager | Meeting Agenda Meeting Minutes |
| Procurement Teams | Coordinate with suppliers to ensure procurement alignment | Face to Face Online Meetings Email | Weekly WIth beginning and end of milestone | Procurement Teams | Project Manager | Meeting Agenda Meeting Minutes |

| | | | | | | |
|---|---|---|---|---|---|---|
| Construction Teams | Coordinate decisions related to construction | Face to Face Online Meetings Email | Weekly WIth beginning and end of milestone | Construction Teams | Project Manager | Meeting Agenda Meeting Minutes |
| Maintenance Teams | Coordinate decisions related to maintenance | Face to Face Online Meetings Email | Weekly | Maintenance Teams | Project Manager | Meeting Agenda Meeting Minutes |
| Project Change Meeting | Manage the requests for changing project scope | Face to Face | When needed | Project Team Stakeholder Sponsor | Project Manager | Meeting Agenda Meeting Minutes Change Request |
| Risk Management | Risk identification, risk assessment and contingency plan | Face to Face Online Meeting | Project initiation | Project Team Stakeholders | Project Manager | Risk Response and Contingency Plan |
| Suppliers Meeting | Coordinate decisions related to supply chain | Face to Face Online Meetings Email | Weekly WIth beginning and end of milestone | Suppliers | Project Manage | Meeting Agenda Meeting Minutes |

## 7.3 PMBOK Referenced PM Methods Used in Communications - Discussion

By reading Chapter 10 of PMBOK, we completed the project communications management plan. The process of project communication management in PMBOK is divided into three parts: 10.1 Plan Communications Management, 10.2 Manage Communications and 10.3 Monitor Communications. Our plan includes all the details in the communication, such as communication participants, communication time, communication methods, etc. This ensures effective information transmission.

In addition, we created a Project Information System table based on Chapter 10 of the PMBOK for managing project communications more visually. The table includes the Risk Response and Contingency Plan, Change Request, Quality Assurance Plan, etc. We refer to the tools and techniques for managing communications in section 10.2 of the PMBOK, which helps us to transmit information effectively. On the other hand,

Through reading section 10.2.1.1 of the PMBOK, we created the Project Communication Plan. This plan describes how to plan, structure, monitor and control project communications, which encourages communication between project teams to enhance teamwork.

# 8. Resources Management Plan

## 8.1 Project Organisation Chart

*Figure 8.1 Project Organisation Chart*

## 8.2 Roles and Responsibilities Table RACI Chart

| roles | responsibilities |
|---|---|
| project director | · Overall oversight of all WBS elements<br>· Final approval of Project System Design<br>· Strategic oversight of budget management<br>· Final sign-off on all major deliverables |

| | |
|---|---|
| | · Project completion approval |
| Project Manager | · Overall management of all WBS elements<br>· Coordinate Project System Design activities<br>· Oversee Mechanical Plant procurement<br>· Manage Electrical Infrastructure implementation<br>· Supervise Civil Work execution<br>· Direct Monitoring & Control setup<br>· Oversee Testing and Commissioning<br>· Ensure proper documentation |
| Engineering Manager | · Site Survey and Study Coordination<br>· Geotechnical Report Oversight<br>· PV Cell Design Management<br>· Electrical Plant Design Supervision<br>· Control & Instrumentation Specification oversight |
| Lead Civil Engineer | · Earthworks design<br>· PV Module Foundations design<br>· Switchyard Foundations design<br>· Reticulation Cabling Trenches Design<br>· Roads design<br>· Drainage design<br>· Site Buildings Design |
| Civil Design Engineers | 1. Earthworks Design Engineer:<br>· Land clearing design<br>· Land Leveling Design<br>2. Foundation Design Engineer:<br>· Steel Support Racks Foundation design<br>· PV Modules foundation design<br>· Switchyard Foundation design |
| Lead Electrical Engineer | · Cabling layout and Switchyard design<br>· Inverters arrangement design |

| | |
|---|---|
| | · Grid connection design<br>· Electrical infrastructure oversight |
| Control Systems Lead | · Instruments specification<br>· Site Controls Design<br>· Telemetry system design<br>· UNSW Campus Control Room design |
| Construction Manager | · Overall management of WBS 1.4 implementation<br>· Coordinate all civil construction activities<br>· Oversee mechanical installation<br>· Supervise electrical installation |
| Site Manager | · Day-to-day management of site activities<br>· Coordinate earthworks<br>· Manage foundation construction<br>· Oversee infrastructure installation |
| Quality Manager | · Quality oversight across all WBS elements<br>· Testing and commissioning quality<br>· Documentation quality control |
| HSE Manager（Health,Safety,Environment） | · Safety oversight across all WBS elements<br>· Site safety management<br>· Installation safety procedures<br>· Testing safety protocols |
| Commissioning Manager | · Performance Testing<br>· Security Inspections<br>· Operation Test<br>· Stability Test |
| Documentation Manager | · Monitoring Training Documents<br>· Equipment Handbooks<br>· Surveillance Training Files |

| | |
|---|---|
| | · User Guides |

## 8.3 Position Descriptions

### Project Director

**Reports to:** UNSW Senior Management

**Responsibility Overview:** Responsible for the overall strategic direction and governance of the program, ensuring that the program is aligned with the UNSW 2025 strategic goals and has final approval authority for key decisions, major changes, and budget expenditures.

**WBS Responsibilities:**

· Overall oversight of all WBS elements (1.1 - 1.7)

· Final approval of Project System Design (1.1)

· Strategic oversight of budget management

· Final sign-off on all major deliverables

· Project completion approval (1.7)

### Project Manager

**Reports to:** Project Director

**Responsibility Overview:** Responsible for the overall execution and day-to-day management of the project, coordinating the teams' work to ensure that the project is completed within budget and schedule, as well as managing project risks and stakeholder relationships.

**WBS Responsibilities:**

· Overall management of all WBS elements (1.1 - 1.7)

· Coordinate Project System Design activities (1.1)

· Oversee Mechanical Plant procurement (1.2)

· Manage Electrical Infrastructure implementation (1.3)

· Supervise Civil Work execution (1.4)

· Direct Monitoring & Control setup (1.5)

· Oversee Testing and Commissioning (1.6)

· Ensure proper documentation (1.7)

### Engineering Manager

**Reports to:** Project Manager

**Responsibility Overview:** Leads all technical aspects of the project, manages the engineering team, ensures that the design meets technical standards, and coordinates the resolution of various technical issues.

**WBS Responsibilities:**

· Primary responsibility for WBS 1.1 (Project System Design)

- 1.1.1 Site Survey and Study Coordination
- 1.1.2 Geotechnical Report Oversight
- 1.1.3 PV Cell Design Management
- 1.1.4 Electrical Plant Design Supervision
- 1.1.5 Control & Instrumentation Specification oversight

## Lead Civil Engineer

**Reports to:** Engineering Manager

**Responsibility Overview:** Responsible for all civil engineering design work for the project, ensuring the integrity of the structural design and providing technical guidance to the construction team.

**WBS Responsibilities:**

- Primary focus on WBS 1.4 (Civil Work)
- 1.4.1 Earthworks design
- 1.4.2 PV Module Foundations design
- 1.4.3 Switchyard Foundations design
- 1.4.4 Reticulation Cabling Trenches Design
- 1.4.5 Roads design
- 1.4.6 Drainage design
- 1.4.7 Site Buildings Design

## Civil Design Engineers

**Reports to:** Lead Civil Engineer

**Responsibility Overview:** Responsible for specific civil engineering design work, including calculations, drawings and preparation of technical specifications, and providing technical support during the construction phase.

**WBS Responsibilities:**

1. Earthworks Design Engineer
- 1.4.1.1 Land clearing design
- 1.4.1.2 Land Leveling Design
1. Foundation Design Engineer
- 1.4.2.1 Steel Support Racks Foundation design
- 1.4.2.2 PV Modules foundation design
- 1.4.3.1 Switchyard Foundation design

## Lead Electrical Engineer

**Reports to:** Engineering Manager

**Responsibility Overview:** Responsible for all electrical system design and implementation for the project, ensuring reliability of the electrical system design and coordination of grid access requirements.

**WBS Responsibilities:**

- Primary focus on WBS 1.1.4 and 1.3
- 1.1.4.1 Cabling layout and Switchyard design
- 1.1.4.2 Inverters arrangement design
- 1.1.4.3 Grid connection design
- 1.3.1-3 Electrical infrastructure oversight

## Control Systems Lead

**Reports to:** Engineering Manager

**Responsibility Overview:** Responsible for the control system and monitoring system design of the project to ensure system integration and remote monitoring functions are realized.

**WBS Responsibilities:**

- Primary focus on WBS 1.5 (Monitoring & Control)
- 1.5.1 Instruments specification
- 1.5.2 Site Controls Design
- 1.5.3 Telemetry system design
- 1.5.4 UNSW Campus Control Room design

## Construction Manager

**Reports to:** Project Manager

**Responsibility Overview:** Responsible for on-site construction management, ensuring project quality, safety and progress, and coordinating the work of various construction teams.

**WBS Responsibilities:**

- Overall management of WBS 1.4 implementation
- Coordinate all civil construction activities
- Oversee mechanical installation (1.2)
- Supervise electrical installation (1.3)

## Site Manager

**Reports to:** Construction Manager

**Responsibility Overview:** Responsible for the day-to-day operation and management of the construction site to ensure that all construction activities are carried out in an orderly manner and to deal with site emergencies.

**WBS Responsibilities:**

- Day-to-day management of site activities
- Coordinate earthworks (1.4.1)
- Manage foundation construction (1.4.2, 1.4.3)
- Oversee infrastructure installation (1.4.4 - 1.4.7)

## Quality Manager

**Reports to:** Project Director

**Responsibility Overview:** Responsible for the establishment and implementation of the project quality management system to ensure that all aspects of the project meet quality standards.

**WBS Responsibilities:**

· Quality oversight across all WBS elements

· Testing and commissioning quality (1.6)

· Documentation quality control (1.7)

## HSE Manager（Health,Safety,Environment）

**Reports to:** Project Manager

**Responsibility Overview:** Responsible for the health, safety and environmental management of the project to ensure construction safety and prevent accidents.

**WBS Responsibilities:**

· Safety oversight across all WBS elements

· Site safety management (1.4)

· Installation safety procedures (1.2, 1.3)

· Testing safety protocols (1.6)

## Commissioning Manager

**Reports to:** Project Manager

**Responsibility Overview:** Responsible for the commissioning and acceptance of the project to ensure that the system functions meet the design requirements.

**WBS Responsibilities:**

· Primary focus on WBS 1.6

· 1.6.1 Performance Testing

· 1.6.2 Security Inspections

· 1.6.3 Operation Test

· 1.6.4 Stability Test

## Documentation Manager

**Reports to:** Project Manager

**Responsibility Overview:** Responsible for the management and preparation of project documentation, ensuring the integrity of all technical documentation and training materials.

**WBS Responsibilities:**

· Primary focus on WBS 1.7

· 1.7.1 Monitoring Training Documents

· 1.7.2 Equipment Handbooks

· 1.7.3 Surveillance Training Files

· 1.7.4 User Guides

## 8.4 Project Staffing Strategy

The project staffing strategy is to ensure that the project team has access to appropriate professionals at all stages to support the smooth progress of the project and meet the requirements of schedule, quality and cost control. The following is a detailed strategy for staffing:

1. **Staffing Goal**
   The goal of project staffing is to ensure that the right people and resources are in place at different stages of the project based on the project's work breakdown structure (WBS) and schedule. This means that in the design, implementation, testing, and delivery stages of the project, appropriate personnel need to be assigned based on the complexity and workload of specific tasks.
2. **Staffing Requirements Analysis**
   Before staffing, it is first necessary to clarify the staffing requirements for each project stage. By analyzing each task in the WBS, determine which positions require specialized technical support, management, procurement, construction, and testing personnel.
   a. **Design Phase:** In the initial stage of the project, senior technicians and project managers are required to be responsible for design and planning.
   b. **Implementation Phase:** During the implementation process, the participation of technical experts, engineers, procurement personnel, logistics personnel, and on-site construction personnel is essential.
   c. **Testing and Delivery Phase:** This phase requires professional test engineers, quality control personnel, and document writers to ensure that the project meets delivery standards.
3. **Staffing Schedule**
   a. **Initial stage (the first two quarters of the project launch):**
   **Main positions:** project manager, technical expert, project engineer, procurement personnel.
   **Goal:** Ensure the smooth progress of project design and planning.
   b. **Mid-term (3rd to 4th quarter):**
   **Main positions:** construction personnel, technical support personnel, quality control personnel, test engineers.
   **Goal:** Support the implementation and installation of the project to ensure that the project is completed on time.
   c. **Late (last two quarters):**
   **Main positions:** testers, quality inspectors, document writers.
   **Goal:** Perform performance testing and acceptance of the project, monitor the construction of the system, and write the final delivery document.
4. **Personnel evaluation and adjustment**
   The performance of the project team will be evaluated regularly to ensure the effectiveness and efficiency of staffing. Through the evaluation of personnel performance, the team configuration can be adjusted in time:
   a. **Regular project review:** monthly review meetings of project personnel are held to evaluate whether the current staffing meets the project requirements.

   b. **Flexible adjustment:** timely adjust the team size and personnel division according to the changes in project requirements to ensure that there are no bottlenecks or waste in personnel in the project.

5. **Summary**
   The core of the project staffing strategy is to reasonably allocate and manage team resources according to the needs of different stages of the project. While ensuring that key personnel are in place in a timely manner, flexible use of external resources and training to improve the skills of internal employees can ensure the smooth completion of the project. Regular personnel assessment and risk management mechanisms will ensure the efficient operation of the project team.

## 8.5 PMBOK Referenced PM Methods Used in Resources Plan – Discussion

In Chapter 9 of PMBOK, it states that Project Resource Management includes the processes of identifying, acquiring, and managing the resources needed to complete the project. These processes help ensure that the project manager and project team have the right resources in the right place at the right time.

The process of project resource management includes:

1. Resource Management Plan: The process of defining how to estimate, acquire, manage, and use material and team resources.
2. Estimate Activity Resources: The process of estimating the types and quantities of team resources and materials, equipment, and supplies needed to complete project work.
3. Acquire Resources: The process of acquiring team members, facilities, equipment, materials, supplies, and other resources needed to complete project work.
4. Team Development: The process of enhancing project performance by improving capabilities, enhancing team member interactions, and optimizing the team environment.
5. Team Management: The process of tracking team member performance, providing feedback, resolving issues, and managing team changes to optimize project performance.
6. Resource Control: The process of ensuring that physical resources assigned to the project are in place as scheduled, monitoring the variance between planned and actual resource use, and taking corrective action when necessary.

**Key Concepts For Project Resource Management:**

The project team consists of individuals with designated roles and responsibilities who work together to achieve shared project goals. The project manager should invest appropriate efforts to acquire, manage, motivate, and enable project team members. Despite the specific roles and responsibilities assigned to project team members, it is beneficial to involve all team members in project planning and decision making. The participation of team members in the planning phase adds their expertise to the process and strengthens their commitment to the project.

The project manager should be both a leader and manager of the project team. In addition to project management activities such as initiating, planning, executing, monitoring and controlling, and closing each project phase, the project manager is also responsible for the team to form an effective working team. The project manager should be aware of various factors that affect the team, such as:

1. Team environment
2. Geographic location of team members
3. Communication with stakeholders
4. Organizational change management
5. Internal and external politics
6. Cultural issues and organizational uniqueness
7. Other factors that may change project performance

**Discussion and Conclusion:**

Project resource management is one of the key links to ensure project success. It includes identifying, acquiring, allocating and managing all resources needed to complete the project, including manpower, materials, equipment and facilities. Through accurate resource planning, resource estimation and acquisition, projects can ensure that the right resources are in the right place at the right time, thereby improving efficiency and controlling costs. Resource management also covers team development and management, ensuring that team members have the necessary skills, promoting collaboration and communication among teams, and thus optimizing project performance. In addition, project managers need to continuously monitor the use of resources, avoid excess or shortage of resources, and take corrective measures in a timely manner to ensure that the project proceeds smoothly as planned and ultimately completes the project goals on time and within budget.